STUDENT ASSIGNMENT MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

CB23F35 –
LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY
COMPUTER SCIENCE AND BUSINESS SYSTEMS



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NOVEMBER 2026

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BONAFIDE CERTIFICATE

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I, Pavitha M (241401063), hereby declare that the project report titled “Student Assignment Management System”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

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ACKNOWLEDGEMENT

I express my sincere gratitude to our Chairman **Mr. S. Meganathan, B.E., F.I.E.,** Chairperson **Dr. Thangam Meganathan, Ph.D.,** and Vice-Chairman **Mr. Abhay Shankar Meganathan, B.E., M.S.,** for providing the infrastructure and facilities that made this work possible.

I am grateful to our Principal, **Dr. S. N. Murugesan, M.E., Ph.D.,** Rajalakshmi Engineering College, for the constant encouragement extended to students to take up industry-relevant learning.

I thank **Dr. K. R. Sowmia, Ph.D.,** Associate Professor & Head, Department of Computer Science and Business Systems, for the guidance and support provided throughout the course.

I owe my sincere thanks to my course faculty, **M. L. Alphin Ezhil Manuel,** Assistant Professor (SG), Department of Computer Science and Business Systems, for the valuable guidance, reviews and feedback that shaped this application and report.

I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

PAVITHA M

ABSTRACT

In many colleges, assignments are announced in class or on messaging groups, submissions are collected on paper or by e-mail, and marks are kept in separate registers. Teachers cannot easily see who has submitted, who is late and who is still pending, and students do not get a clear record of their submission status, marks and feedback. The result is repeated enquiries, inconsistent treatment of late work and no single record from which progress can be reviewed.

Student Assignment Management System is a low-code application built on Zoho Creator to replace this manual practice with one structured system. Low-code suits the problem because assignment management is a data-entry and workflow task that the platform's forms, reports and dashboard components cover directly. The application contains four linked forms – Student Details, Subject Details, Assignment Details and Submission Details – joined by lookup fields. Reports list the students, subjects, assignments and submissions, and a dashboard page gives student and subject counts, a pie chart and shortcut buttons. Two Deluge scripts validate a submission and set its status to Late or Submitted by comparing the submission date with the assignment's due date. Five form workflows are enabled: four on the Submission Details form, which also send an e-mail to the administrator when a submission is recorded and an e-mail to the student when the status is updated, and one on the Assignment Details form, which is configured to create a Zoho Calendar event. Access is governed by a role hierarchy of CEO, Admin, Teacher and Student, three permission profiles (Admin, Teacher and Student) and two data sharing rules.

The application was tested through its live interface, and the delivery of the administrator e-mail and of the student e-mail was confirmed in the recipient mailbox. Version 2.0 was published from the Development environment to Production on 1 November 2026. The report also records the defects found in the first build and the limitations that remain, which are listed as future enhancements.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Assignment Management; Role-Based Access Control

PROJECT AT A GLANCE

Attribute	Details
Project Title	Student Assignment Management System
Application Domain	Academic administration: assignments, submissions and evaluation in a college
Student Name & Register No.	PAVITHA M – 241401063
Zoho Creator Edition	Trial edition
Application Name (Link Name)	Student Assignment Management System
Published / Live URL	https://pavitham2024csbsraja1141.zohocreatorportal.in (portal)
Number of Forms / Subforms	4 (Student Details, Subject Details, Assignment Details, Submission Details) / 0
Number of Reports (by type)	6 report entries: All Student Details, student details report, subject details report, Assignment Details Report, All Submission Details, submission details report (list reports)
Number of Pages / Dashboards	1 (dashboard with student and subject count tiles, a pie chart, two shortcut buttons and a status panel)
Number of Workflows	Form: 5 (4 on Submission Details, 1 on Assignment Details) · Schedule, Approval, Blueprint, Batch: none
Deluge Functions / Custom Actions	Deluge scripts in the form workflows (validation, late-submission check, e-mail notifications, calendar event); no stand-alone function
Integrations	Creator e-mail notifications to the administrator and the student; Zoho Calendar (workflow on Assignment Details)
Roles & Profiles Defined	Roles: CEO, Admin, Teacher, Student · Permission profiles: Admin_Permission, Teacher_Permission, Student_Permission (with the default Read and Write profiles) · Data sharing rules: 2
Environments Used	Development and Production
Date of Production Deployment	1 November 2026, 15:14 IST (version 2.0)

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CEO	Chief Executive Officer
ER	Entity–Relationship
FR	Functional Requirement
IST	Indian Standard Time
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
TC	Test Case
UAT	User Acceptance Testing
UC	Use Case
UI	User Interface
URL	Uniform Resource Locator
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Colleges set assignments in almost every subject. In most departments the process is informal: a teacher announces an assignment in class or on a messaging group, students hand in work on paper or send it by e-mail, and the teacher records who has submitted and the marks in a register or a personal spreadsheet. Each step is done by hand. Students have no dependable record of their submission status, and teachers have no consolidated view of how many students have submitted, how many are late and how many are still pending. The result is repeated enquiries, inconsistent treatment of late work and marks that are kept apart from the submissions they belong to.

Student Assignment Management System is a low-code application built on Zoho Creator to replace this manual practice with one structured system. Student and subject records are kept in forms. A teacher creates an assignment for a subject with an assigned date, a due date and maximum marks. A submission is recorded against a student and an assignment; the application validates it, decides automatically whether it is Late or Submitted by comparing the dates, and e-mails the administrator. When the teacher updates the status, for example to Evaluated after marking, the student receives an e-mail. Reports and a dashboard present the records, and roles with permission profiles control who may see and change them.

The application was first built as a set of linked forms and reports and was then reviewed against its own test evidence. The findings of that review shaped the present version, which adds the validation and late-submission scripts, the notification workflows, the role hierarchy with permission profiles and data sharing rules, and a dashboard. Version 2.0 of the application, named Student Assignment Management System in the Creator console, was published to the Production environment on 1 November 2026 (Figure 5.1).

1.2 Problem Statement

Manual handling of assignments gives neither teachers nor students reliable information about submission status. A low-cost digital solution is required that stores student, subject and assignment details, records every submission with its date, status, marks and feedback, decides automatically whether a submission is late, informs the administrator and the student by e-mail, controls access by role and presents the position on a single dashboard. The main pain points are:

- Student, subject and assignment details are kept in different places, so a teacher must search several records to find one student’s work.
- Teachers cannot see, at a glance, how many submissions are pending, late or evaluated for an assignment.
- Late submissions are judged by hand and therefore inconsistently.

- Students are not told when the status of a submission changes, and marks and feedback are not stored with the submission they belong to.

1.3 Objectives

The project set out to achieve the following objectives:

- To design a data model of four linked forms – Student Details, Subject Details, Assignment Details and Submission Details – with lookup relationships from each assignment to its subject and from each submission to its student and assignment.
- To validate each submission on saving and to set its status to Late or Submitted automatically by comparing the submission date with the assignment's due date, using Deluge scripts in form workflows.
- To notify the administrator by e-mail when a submission is recorded and to notify the student by e-mail when the submission status is updated.
- To configure a Zoho Calendar workflow that creates an event for each assignment.
- To provide reports and a dashboard that show students, subjects, assignments and submissions, with counts and a chart.
- To control access through a role hierarchy (CEO, Admin, Teacher, Student), three permission profiles and data sharing rules.
- To publish the application from the Development environment to Production.

1.4 Scope of the Project

In scope. The implemented application contains four forms (Student Details, Subject Details, Assignment Details and Submission Details), six report entries, one dashboard page, five form workflows (four on Submission Details and one on Assignment Details), two Deluge scripts that can be read in the editor, e-mail notifications to the administrator and the student, a Zoho Calendar workflow, a role hierarchy of CEO, Admin, Teacher and Student, three permission profiles and two data sharing rules. The application is a general-purpose assignment tracker for a college department and is demonstrated with sample students and subjects such as ai, sduml, cnt and low code.

Out of scope. The following were considered but are not part of the delivered application and are treated in this report as future enhancements: upload of the assignment file with a submission, reminder e-mails before the due date, automatic calculation of marks or grades, plagiarism checking, SMS notification, a native mobile application configuration and integration with third-party services through REST API calls.

1.5 Justification for a Low-Code Approach

An assignment tracker is a data-entry and workflow application: a few forms, a handful of reports, simple date comparisons and notifications. These are exactly the building blocks that a low-code application platform provides as configurable components. On Zoho Creator the forms,

their lookup relationships, the reports and the dashboard page were assembled with the form builder, report builder and page builder, and only the validation, the late-submission check and the notification messages required scripting in Deluge. Hosting, authentication, user management, responsive rendering and record storage are supplied by the platform, so the project effort could be concentrated on the assignment logic rather than on infrastructure. Published studies of low-code practice report that drag-and-drop development and out-of-the-box components speed up development and that low-code is most favoured for process- and workflow-centred applications, although practitioners also report limits in customisation [4], [5].

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The existing arrangement for assignments in most college departments is manual. A teacher announces an assignment in class or on a messaging group. Students note the due date for themselves and hand in the work on paper or send it by e-mail. The teacher notes each submission in a register or a personal spreadsheet, decides by hand whether it is late, marks the work and records the marks. Students learn their marks only when they ask or when the marks are announced. Figure 2.1 summarises this commonly followed practice. The description is a general account of manual assignment handling and is not a survey of one named institution.

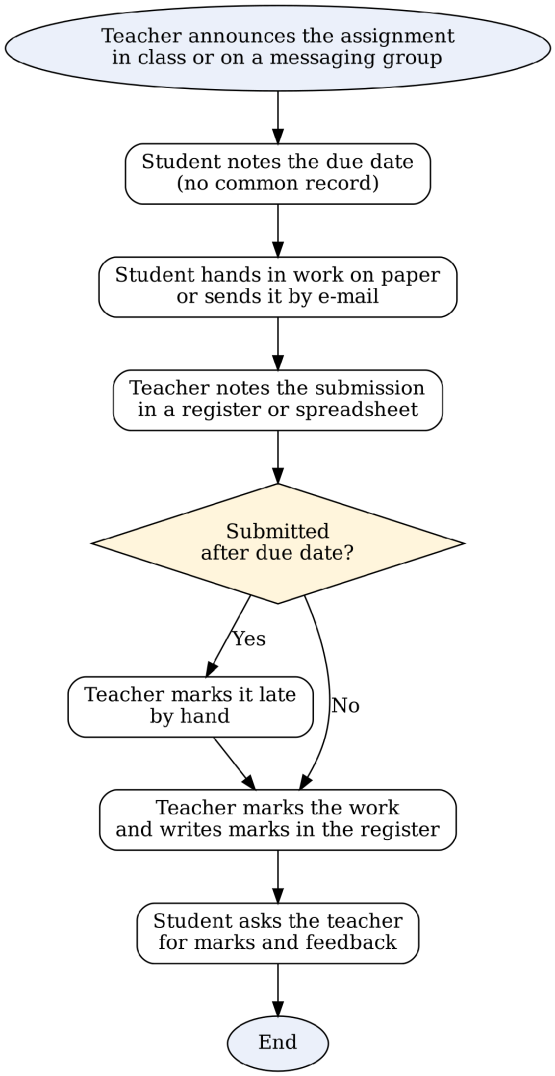


Figure 2.1: As-Is Process Flow of Manual Assignment Handling in a College Department

2.2 Limitations of the Existing System

- **No consolidated view.** Teachers cannot see the number of submitted, late and pending submissions per assignment without counting entries in a register.
- **Scattered records.** Student, subject and assignment details live in different registers and files, so one student's work is hard to trace.
- **Inconsistent late rules.** Whether a submission is late is judged by hand, so identical cases may be treated differently.
- **No automated alerts.** Neither the administrator nor the student is notified when a submission is recorded or when its status changes.
- **Marks kept apart.** Marks and feedback are not stored with the submission, so they are easily lost or disputed.
- **No access control.** Anyone who holds a register or spreadsheet can read or change it; there is no separation between what an administrator, a teacher and a student may see.

2.3 Low-Code versus Traditional Development

Table 2.1 compares a traditional pro-code implementation of the assignment manager with the low-code implementation on Zoho Creator. The comparison is qualitative; no development-time or cost measurements were recorded during the project, so none are claimed.

Table 2.1: Comparison of Low-Code and Traditional Development for the Student Assignment Management System

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Database design, back-end services, user interface and deployment pipeline must each be built and integrated.	Forms, reports and the dashboard are assembled in builders; Deluge was needed only for validation, the late-submission check and notification logic.
Skills required	Programming in a front-end and a back-end language, SQL, hosting and security administration.	Platform familiarity and basic Deluge scripting.
Hosting and infrastructure	Servers or cloud services must be provisioned, secured and maintained.	Hosted and maintained by the platform.
Access for users	A web site with log-in pages and a responsive front end must be developed.	Users sign in to the hosted application or portal, and the platform renders applications on desktop, tablet and phone previews.
Notifications and integration	E-mail servers, calendar APIs, authentication tokens and error handling must be coded.	E-mail actions are configured in workflows, and the calendar task is written in Deluge on top of a platform connection.
Security and access control	Authentication, sessions, role logic and record sharing must be implemented in code.	Users, roles, permission profiles and data sharing rules are configured in settings.
Maintenance	Changes require code changes, testing and redeployment.	Changes are made in the builder and published from Development to Production.

2.4 Literature Survey

Table 2.2 summarises the sources reviewed. Two are empirical studies of low-code practice; three are the vendor documentation sets for the platform services on which the application depends. Only findings that could be confirmed from the cited pages are recorded.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Y. Luo, P. Liang, C. Wang, M. Shahin and J. Zhan, 2021 [4]	Characteristics and Challenges of Low-Code Development: The Practitioners' Perspective	Analysis of Stack Overflow and Reddit posts found that drag-and-drop interfaces and out-of-the-box components make low-code platforms easy to learn and speed up development, that they are favoured for automated processes and workflows, and that practitioners hold conflicting views of their advantages and disadvantages.	Supports the choice of a low-code platform for a workflow-oriented assignment application and motivates the limitations discussion in Section 8.4.
2	M. A. Al Alamin, S. Malakar, G. Uddin, S. Afroz, T. B. Haider and A. Iqbal, 2021 [5]	An Empirical Study of Developer Discussions on Low-Code Software Development Challenges	Stack Overflow posts on nine low-code platforms identified 13 discussion topics in four categories; customisation was the largest category, and dynamic event handling was both the most popular and the hardest topic to obtain accepted answers for.	Highlights event handling and customisation as risk areas, which corresponds to the form-event workflows and Deluge scripts used in this application.
3	Zoho Corporation [1]	Zoho Creator Resource Center	Central documentation for building applications, using Deluge, managing users and assigning roles and permissions.	Reference for forms, reports, pages, workflows and user permissions.
4	Zoho Corporation [2]	Introduction to Deluge (Zoho Deluge Help)	Describes Deluge as Zoho's scripting language used to manipulate data, for example by creating or updating records.	Reference for the validation, late-submission and notification scripts.
5	Zoho Corporation [3]	Zoho Cliq Help Center	Documentation for Zoho Cliq, the team chat service, including its integrations with other applications.	Background reference only; Zoho Cliq is not used in this application.

2.5 Emerging Trends in Low-Code Development

Three general trends are relevant to this project. First, low-code platforms increasingly embed AI assistance in their builders; the Zia assistant icon is visible in the Zoho Creator builder screens captured for this project, although no AI feature was used in this application. Second, platforms emphasise integration with the surrounding service ecosystem, which this application exercises through e-mail actions and a Zoho Calendar workflow. Third, as more business users build applications, governance concerns such as role design, data sharing, data privacy settings and environment management become more important; this application addresses these with a role hierarchy, permission profiles, sharing rules and a Development-to-Production release. The empirical literature also records that customisation limits and event handling remain recurring

difficulties for developers [4], [5]. These trends are stated qualitatively and are not backed by market figures.

2.6 Proposed System

The proposed system is the Student Assignment Management System application. Students and subjects are entered once in their forms. A teacher creates an assignment by choosing a subject on the Assignment Details form and entering the title, description, assigned date, due date and maximum marks. A submission is recorded on the Submission Details form by choosing a student and an assignment from the lookup lists and entering the submission date, the status, the marks obtained and feedback. On saving, the application checks that the assignment, the student and the status are filled, decides whether the submission is Late or Submitted by comparing the submission date with the due date, and e-mails the administrator. When the status is updated the student receives an e-mail. Records are visible in reports and on a dashboard, and access is controlled through a role hierarchy, permission profiles and data sharing rules. The proposed system therefore removes the manual register, gives teachers a single view of assignments and submissions, gives students a record of their status and gives the administrator an organised record of academic activity.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

The roles below correspond to the roles and permission profiles defined in the application settings.

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Role “Admin” in the role hierarchy beneath CEO; permission profile “Admin_Permission”.	Complete access to the application: adds, views, edits and deletes student details, subject details, assignments and submissions; accesses reports and dashboards; manages users and settings.
Teacher	Role “Teacher” beneath CEO; permission profile “Teacher_Permission”.	Views student and subject details; creates and manages assignments; views student submissions; updates submission status; evaluates assignments; enters marks and feedback.
Student	Role “Student” beneath Teacher; permission profile “Student_Permission”.	Views available assignments; submits assignments; checks submission status; views marks and feedback; accesses relevant reports.

3.2 Functional Requirements

Table 3.2 lists the functional requirements. FR-01 to FR-12 correspond to features implemented in the application; FR-13 and FR-14 are proposed requirements that have not been implemented.

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall store student details: Student ID, Student Name, Email Address, Department (mandatory) and Phone number.	Admin	High
FR-02	The system shall store subject details: Subject ID, Subject Name, Faculty Name and Semester.	Admin	High
FR-03	The system shall allow assignments to be created with a title, a subject chosen from the subject list, a description, an assigned date, a due date and maximum marks.	Teacher, Admin	High
FR-04	The system shall record a submission for a student and an assignment, chosen from lookup lists, with submission date, submission status, marks obtained and feedback.	Student, Teacher, Admin	High
FR-05	The system shall reject a submission in which the assignment, the student or the submission status is empty and shall show a message for each.	System	High
FR-06	The system shall set the submission status to Late when the submission date is after the due date of the assignment, and to Submitted otherwise.	System	High
FR-07	The system shall e-mail the administrator the student, assignment and status when a submission is recorded.	System	High

FR ID	Requirement	Role(s)	Priority
FR-08	The system shall e-mail the student the assignment and current status when the submission status is updated.	System	High
FR-09	The system shall run a workflow that creates a Zoho Calendar event for a new assignment.	System	Medium
FR-10	The system shall provide reports of students, subjects, assignments and submissions.	Admin, Teacher	High
FR-11	The system shall provide a dashboard with counts of students and subjects, a chart and shortcut buttons.	Admin, Teacher	Medium
FR-12	The system shall restrict access by role, permission profile and data sharing rule.	Admin	High
FR-13	(Proposed, not implemented) The system shall allow a student to attach the assignment file to a submission.	Student	Medium
FR-14	(Proposed, not implemented) The system shall e-mail students a reminder before an assignment's due date.	System	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports and the dashboard shall open without perceptible delay for the record volumes of one department. No formal timing measurement was recorded.
NFR-02	Security	Users shall access only the forms, reports and pages permitted by their role and permission profile.
NFR-03	Usability	A submission shall be recorded with a single form submission, selecting the student and the assignment from lists; the application shall be usable in a standard web browser.
NFR-04	Availability	The application shall rely on the availability of the Zoho Creator service; no separate hosting is maintained by the project.
NFR-05	Maintainability	Business logic shall be confined to named workflows so that it can be reviewed and changed without redesigning the forms.
NFR-06	Traceability	Each submission shall carry its student, assignment, submission date and status, so that its history can be read from one record.
NFR-07	Reliability of notifications	An e-mail to the administrator shall be produced for every successfully submitted submission, and an e-mail to the student for a status update; their delivery is checked in the mailbox.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (trial edition indicated in the application header), with Development and Production environments
Scripting language	Deluge (Zoho's scripting language for Creator workflows and functions)
Browser	A current desktop web browser; the builder also offers tablet and phone previews

Category	Specification
Other Zoho services	Zoho Calendar (assignment event workflow); Zoho Creator mail service for notifications
Client device	Computer or phone with internet access
Mailbox	A mailbox to receive notifications from the Zoho Creator trial mail address

3.5 Use Case Model

Figure 3.1 shows the use cases of the application. The administrator manages student and subject records, roles and permissions, and views reports and the dashboard. The teacher creates assignments, updates the status, marks and feedback of submissions and views reports. The student submits assignments and views the status, marks and feedback. The administrator also performs the teacher use cases, as indicated by the dashed generalisation arrow, which is consistent with Admin and Teacher both sitting beneath CEO in the role hierarchy. Creating an assignment includes the calendar action, submitting an assignment includes the administrator e-mail, and updating a status includes the student e-mail. Tables 3.5 to 3.7 describe three key use cases in detail.

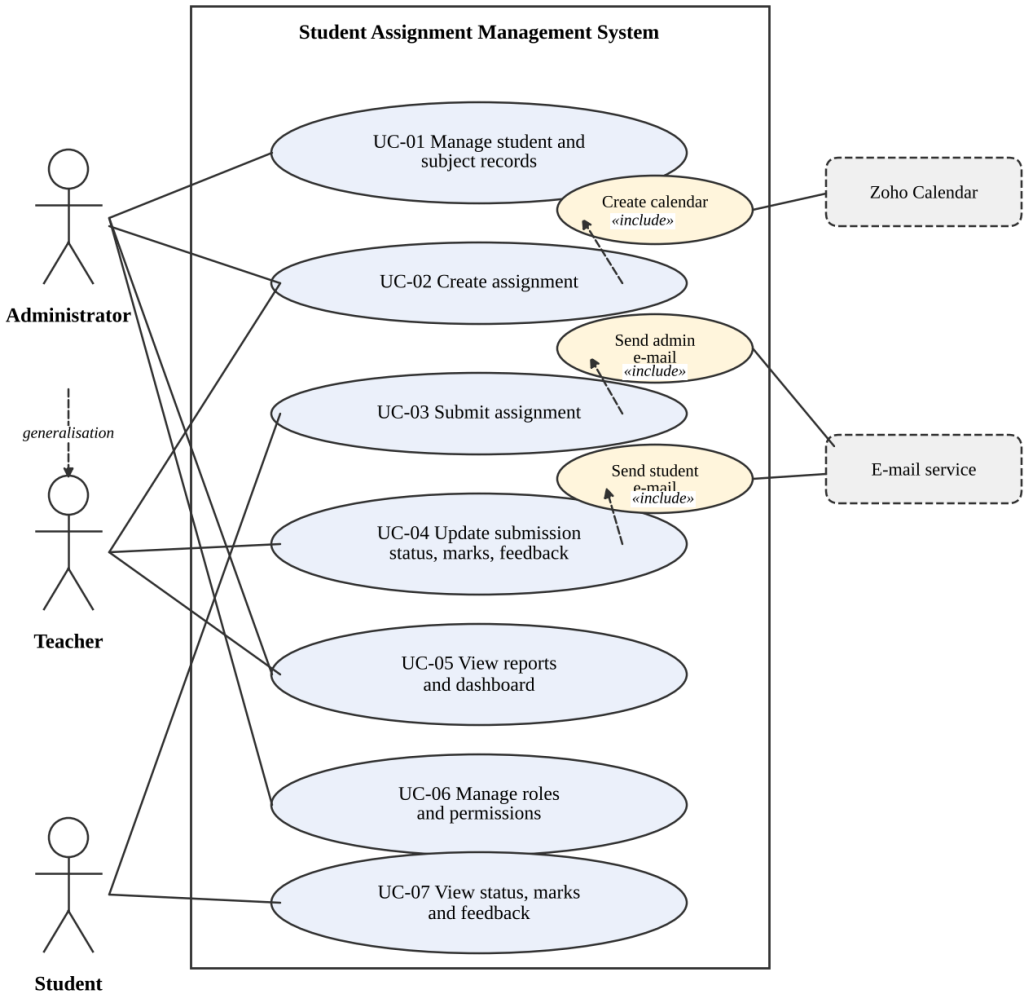


Figure 3.1: Use Case Diagram of the Student Assignment Management System

Table 3.5: Use Case Description – Create Assignment (UC-02)

Use Case ID	UC-02
Use Case Name	Create Assignment
Actor(s)	Teacher (primary); Administrator; Zoho Calendar (secondary)
Pre-condition	At least one Subject Details record exists; the actor has access to the Assignment Details form.
Main Flow	1. The actor opens the Assignment Details form from the Assignment Details menu section. 2. The actor enters the Assignment Title and selects the Subject from the lookup list. 3. The actor enters the Description, Assigned Date, Due Date and Maximum Marks. 4. The actor clicks Submit. 5. The Calendar workflow runs after the record is saved.

Alternate Flow	If the Subject list is empty because no subject has been entered, the actor first adds a record on the Subject Details form.
Post-condition	The assignment appears in the Assignment Details Report and in the Assignment lookup of the Submission Details form.
Related FR	FR-03, FR-09, FR-10

Table 3.6: Use Case Description – Submit Assignment (UC-03)

Use Case ID	UC-03
Use Case Name	Submit Assignment
Actor(s)	Student or Administrator (primary); E-mail service (secondary)
Pre-condition	Student Details and Assignment Details records exist; the actor has access to the Submission Details form.
Main Flow	1. The actor opens the Submission Details form. 2. The actor selects the Student and the Assignment from the lookup lists. 3. The actor enters the Submission Date and selects the Submission Status. 4. The actor clicks Submit. 5. The validation workflow checks that the assignment, student and status are filled. 6. The late-submission workflow compares the submission date with the due date and sets the status to Late or Submitted. 7. After the record is saved, the administrator receives an e-mail with the student, assignment and status.
Alternate Flow	If the assignment, the student or the status is empty, an alert is shown (for example “Please select an assignment.”) and the submission is cancelled.
Post-condition	A Submission Details record appears in the submission reports, and the administrator has been e-mailed.
Related FR	FR-04, FR-05, FR-06, FR-07

Table 3.7: Use Case Description – Update Submission Status (UC-04)

Use Case ID	UC-04
Use Case Name	Update Submission Status
Actor(s)	Teacher (primary); Administrator; E-mail service (secondary)
Pre-condition	A Submission Details record exists; the actor holds the Teacher or Admin profile.
Main Flow	1. The actor opens the submission record from the Submission Details report. 2. The actor changes the Submission Status (for example to Evaluated) and enters the Marks Obtained and Feedback. 3. The actor saves the record. 4. The Student Status Update Notification workflow e-mails the student the assignment and the current status.
Alternate Flow	If a mandatory field is emptied, the validation script cancels the save and shows an alert.
Post-condition	The record shows the new status, marks and feedback, and the student has been notified.
Related FR	FR-04, FR-08

CHAPTER 4
SYSTEM DESIGN

This chapter describes how the Student Assignment Management System is designed, before Chapter 5 shows how it was built. It covers the architecture, the data model, the form specifications, the submission lifecycle and workflows, the navigation and dashboard design, and the access-control design. The design described here is that of the deployed application, the Student Assignment Management System (version 2.0) on Zoho Creator, which was published to production on 1 November 2026.

4.1 System Architecture

The application runs entirely inside Zoho Creator. It has no custom front end, no separate back end and no external database. Figure 4.1 shows the architecture. The administrator, teachers and students use the application through a web browser.

Presentation layer. Four forms (Student Details, Subject Details, Assignment Details and Submission Details), a set of list reports and the dashboard page.

Logic layer. Five form workflows written with Deluge: four on the Submission Details form and one on the Assignment Details form. They validate the submission, set its status, send the notifications and run the calendar action.

Data layer. The records of the four forms, held by Creator and related by lookup fields: an Assignment refers to one Subject, and a Submission refers to one Student and one Assignment.

Notification channels. E-mail is sent to the administrator and to the student from the Zoho Creator mail address of the account, and a calendar event is created through Zoho Calendar.

Access control. A role hierarchy (CEO at the top, Admin and Teacher beneath it and Student beneath Teacher), three permission profiles and two data sharing rules govern what each user can see and do.

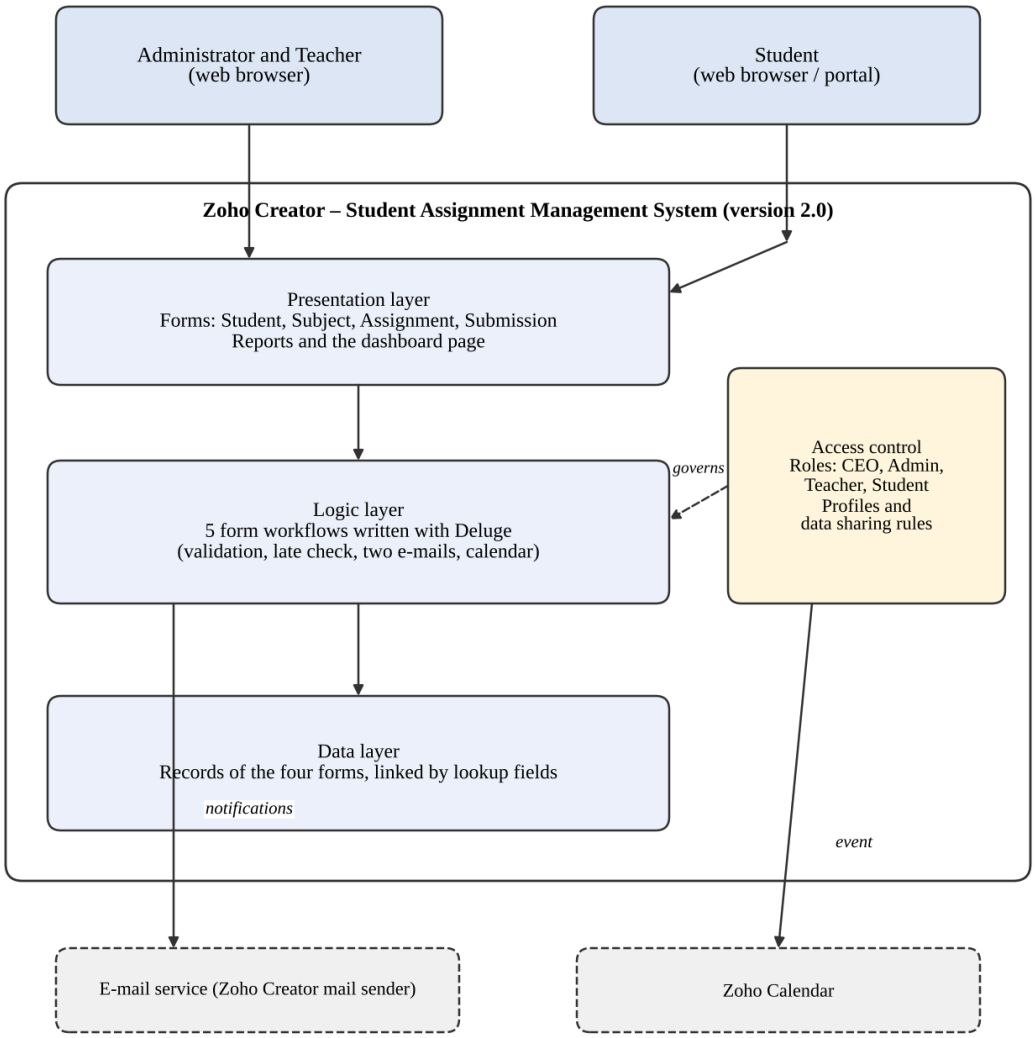


Figure 4.1: System Architecture of the Student Assignment Management System

4.2 Data Model

Zoho Creator stores data as records of forms. A form therefore plays the role that a table plays in a relational database, a field plays the role of a column, and a lookup field plays the role of a foreign key that points to a record of another form. The application uses four forms. Student Details holds one record per student. Subject Details holds one record per subject. Assignment Details holds one record per assignment and carries a Subject lookup, so one subject can have many assignments and each assignment belongs to one subject. Submission Details holds one record per submission and carries a Student lookup and an Assignment lookup, so one student can have many submissions, one assignment can receive many submissions, and each submission belongs to exactly one student and one assignment. Figure 4.2 shows the entities and their relationships, and Table 4.1 summarises them.

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship	Implemented Using
Student Details	Student ID, Student Name, Email Address, Department, Phone number	Submission Details	One-to-Many	Student lookup in Submission Details
Subject Details	Subject ID, Subject Name, Faculty Name, Semester	Assignment Details	One-to-Many	Subject lookup in Assignment Details
Assignment Details	Assignment Title, Subject, Description, Assigned Date, Due Date, Maximum Marks	Subject Details (parent); Submission Details (child)	Many-to-One (subject); One-to-Many (submissions)	Subject lookup; target of the Assignment lookup
Submission Details	Student, Assignment, Submission Date, Submission Status, Marks Obtained, Feedback	Student Details; Assignment Details	Many-to-One	Student and Assignment lookups

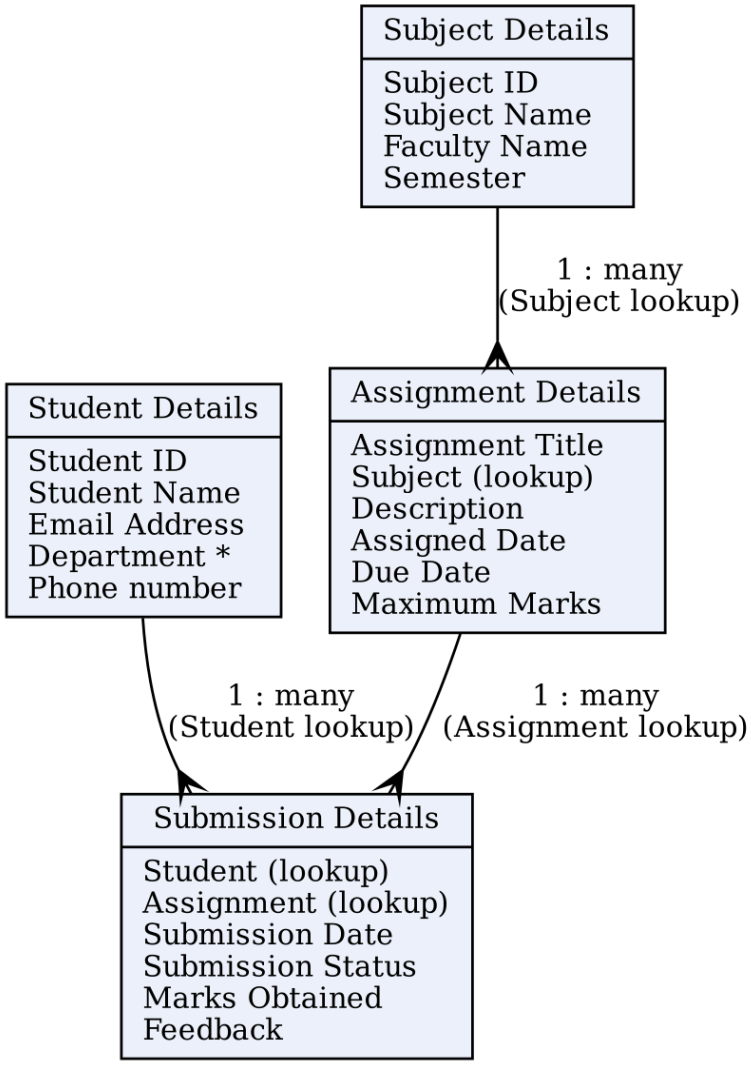


Figure 4.2: Entity–Relationship Diagram of the Student Assignment Management System (* marks a mandatory field)

Data consistency is supported in four ways.

- **Lookups.** The subject on an assignment and the student and assignment on a submission are chosen from existing records, so mistyped names cannot enter the data.
- **Fixed value sets.** Department, Semester and Submission Status are drop-down fields, so only the defined values can be stored.
- **Mandatory fields.** Department is mandatory on Student Details (shown with an asterisk in Figure 5.2).
- **Validation on submission.** The Validate Submission Details workflow runs before a submission is saved and cancels it if the assignment, student or status is empty (Section 4.4).

4.3 Form Design

Tables 4.2 to 4.5 specify the fields of the four forms. Field names, mandatory flags and field types were read from the form builder and the live forms. Where scripts refer to a field, its link name is given in the Remarks column. Field types that are not visible in the screenshots are marked as not shown.

Table 4.2: Field Specification – Student Details Form

Field Label	Field Type	Mandatory	Remarks
Student ID	Single Line (type as shown on the form)	No	Identifies the student; no unique constraint is set (Section 6.4, D-02)
Student Name	Single Line	No	Shown in the Student lookup of Submission Details
Email Address	Email	No	Used to identify the student in the report
Department	Drop Down	Yes	Values seen in the report include Mechanical, Information Technology, Computer Science and Electronics
Phone number	Phone	No	Country-code selector (+91)

Table 4.3: Field Specification – Subject Details Form

Field Label	Field Type	Mandatory	Remarks
Subject ID	Single Line	No	Identifies the subject
Subject Name	Single Line	No	Shown in the Subject lookup of Assignment Details
Faculty Name	Single Line	No	Name of the faculty member for the subject
Semester	Drop Down	No	Semester of the subject

Table 4.4: Field Specification – Assignment Details Form

Field Label	Field Type	Mandatory	Remarks
Assignment Title	Single Line	Not shown	Shown in the Assignment lookup of Submission Details
Subject	Lookup (Subject Details)	Not shown	Values seen: ai, sduml, cnt, low code
Description	Multi Line (type not shown)	Not shown	Free text
Assigned Date	Date	Not shown	Format dd-MMM-yyyy as in the report
Due Date	Date	Not shown	Read by the Check Late Submission script
Maximum Marks	Number	Not shown	Values seen: 75 and 100

Table 4.5: Field Specification – Submission Details Form

Field Label	Field Type	Mandatory	Remarks
Student	Lookup (Student Details)	Checked by script	Drop down showing Student Name; link name Student
Assignment	Lookup (Assignment Details)	Checked by script	Drop down; link name Assignment
Submission Date	Date	No	Placeholder dd-MMM-yyyy; link name Submission_Date
Submission Status	Drop Down	Checked by script	Values seen: Submitted, Evaluated; Late is set by script; link name Submission_Status
Marks Obtained	Number	No	Marks awarded by the teacher
Feedback	Multi Line	No	Teacher’s feedback to the student

4.4 Process and Workflow Design

Submission lifecycle. A submission is entered with a status. When it is saved, the Check Late Submission workflow compares the submission date with the assignment’s due date and stores Late or Submitted. The administrator is then notified. Later changes of status are manual edits by the teacher: after marking, the status is changed to Evaluated and the student is notified. Table 4.6 lists the status values seen in the application, and Figure 4.3 shows the lifecycle. The design contains no scheduled or automatic status change.

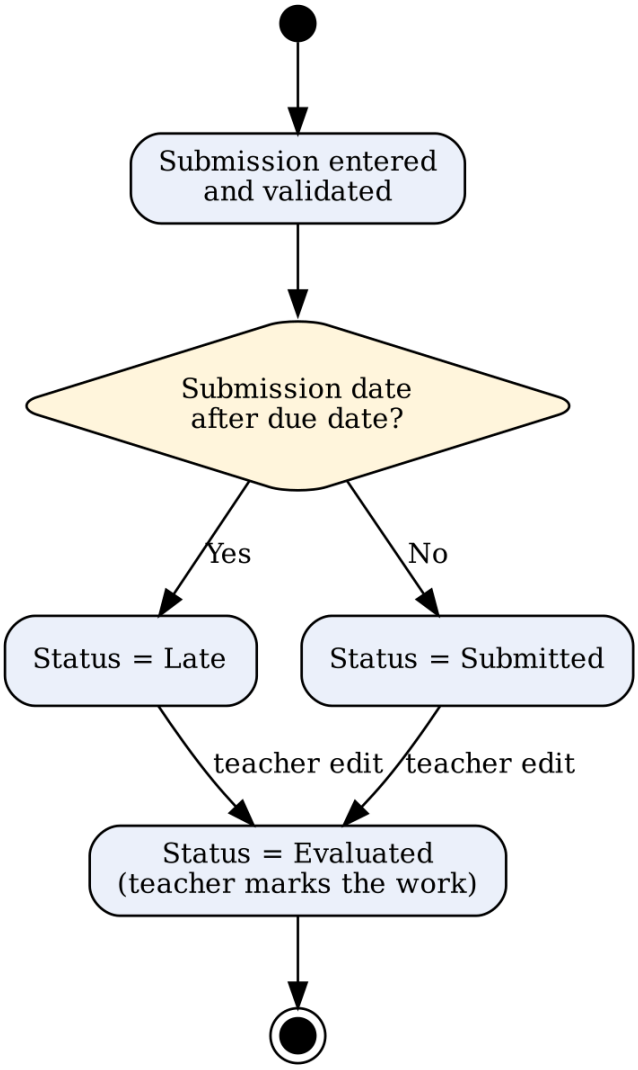


Figure 4.3: Submission Processing Workflow (Lifecycle of a Submission)

Table 4.6: Submission Status Values and Transitions

Status	Entered When	Changed By	Where It Appears
Submitted	The submission date is on or before the due date	Check Late Submission script (automated)	Admin and student e-mails; submission reports
Late	The submission date is after the due date	Check Late Submission script (automated)	Submission reports
Evaluated	The teacher has marked the work	Teacher (manual edit)	Student e-mail (Figure 5.13); admin e-mail (Figure 5.12)

Workflows. The five form workflows, all enabled, are listed in Table 5.6, and Figure 4.4 shows which form event starts each of them. Four are defined on the Submission Details form and one on the Assignment Details form. The event decides when a workflow runs: Validations on form

submission runs before the record is saved, and Successful form submission runs after the record is saved.

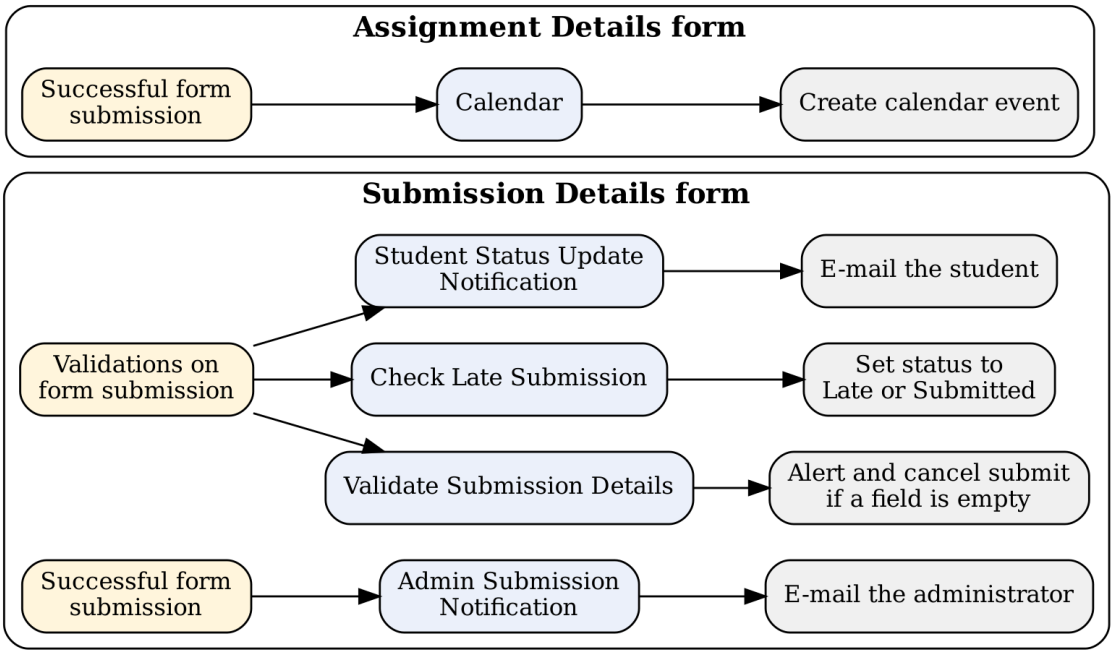


Figure 4.4: Form Events, Workflows and the Actions They Produce

Notification design. Two messages leave the application for a submission. When a submission is recorded, the administrator receives an e-mail titled “New Assignment Submission” that states the student, the assignment and the status. When the status is updated, the student receives an e-mail titled “Assignment Submission Status Updated” that states the assignment ID and the current status. A third action, the Calendar workflow, runs when an assignment is saved.

Late-submission logic. The status is decided by the rule

Submission Status = Late if Submission Date > Due Date; otherwise Submitted

where the due date is read from the Assignment Details record that the submission refers to. Figure 4.5 shows the logic of the Check Late Submission workflow. For example, an assignment due on 02-Nov-2026 that is submitted on 03-Nov-2026 is marked Late, and one submitted on 02-Nov-2026 is marked Submitted.

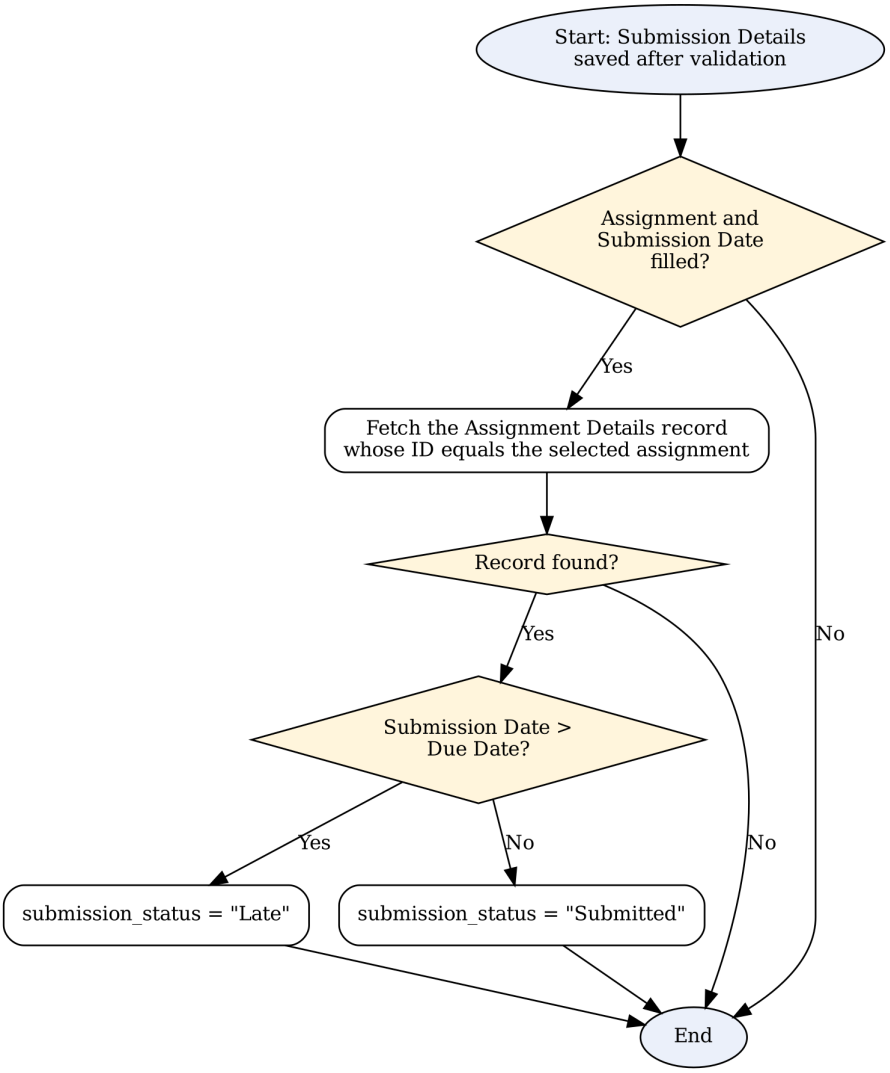


Figure 4.5: Flowchart of the Check Late Submission Workflow

The design has stated limitations. The check runs only when both the assignment and the submission date are filled, so a submission saved without a date keeps the status chosen by hand. The comparison uses dates only and takes no account of a grace period or of the time of day. Marks are entered by the teacher and are not checked against the maximum marks of the assignment. A check that the marks obtained do not exceed the maximum marks is listed as a future enhancement in Chapter 9.

4.5 User Interface and Navigation Design

Figure 4.6 shows the navigation structure of the application. The side menu has four sections and a dashboard. The first section, named Student Details, holds the Student Details form and the reports All Student Details and student details report. A second section, also named Student Details, holds the Subject Details form and the reports All Student Details and subject details report. The Assignment Details section holds the Assignment Details form and the Assignment

Details Report. The Submission Details section holds the Submission Details form and the reports All Submission Details and submission details report. The dashboard is the last menu item.

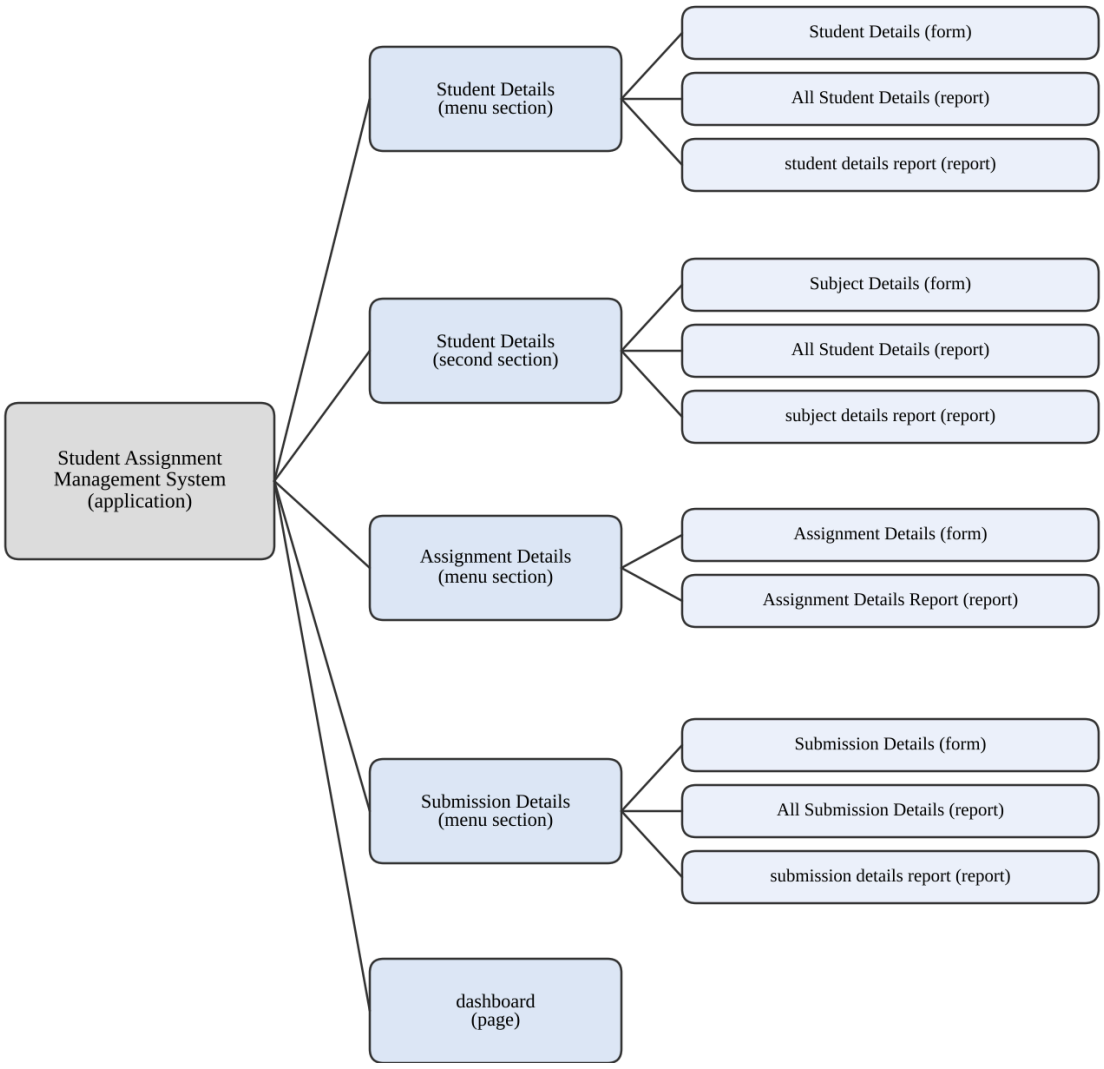


Figure 4.6: Navigation Structure of the Student Assignment Management System

Dashboard. The dashboard is the landing page for the administrator and teachers. It combines a tile with the number of students, a pie chart, a tile with the number of subjects, two buttons ("submit assignment" and "Update submission status") and a status panel with Submitted and Evaluated entries. A person can therefore see the size of the data and open the submission form in one click.

Forms and reports. Each form is a single page of labelled fields, so a user fills it from top to bottom. List reports show records in rows with sortable columns, a search icon and an add button. The interface uses the platform's dark side menu with an indigo accent and can be previewed in desktop, tablet and phone layouts.

4.6 Access Control Design

Access is designed on three levels. Users sign in with their accounts. A role hierarchy places CEO at the top, Admin and Teacher beneath it and Student beneath Teacher. Permission profiles define what each kind of user may do with each form, report and page. Data sharing rules extend record visibility between roles.

Permission profiles. Three profiles are defined for the application – Admin_Permission, Teacher_Permission and Student_Permission – in addition to the default Read and Write profiles. Table 4.7 records the intent of each profile as written in its description. How each profile is set against the individual forms, reports and pages is shown in Chapter 5.

Table 4.7: Permission Profiles and Their Design Intent

Profile	Intended Access	Intended Restriction
Admin_Permission	Complete access to the application: add, view, edit and delete student details, subject details, assignments and submissions	None
Teacher_Permission	View student and subject details, create and manage assignments, view student submissions, update submission status, evaluate assignments, enter marks and provide feedback	No access to user and application settings
Student_Permission	View available assignments, submit assignments, check submission status, view marks and feedback, access relevant reports	No access to other students' records or to administrative features

Role hierarchy and data sharing. In Zoho Creator a role hierarchy governs data visibility between users, whereas a permission profile governs what a user may do. The application has four roles. Two data sharing rules are defined, as shown in Table 5.8: one shares Assignment Details from the Student role to the Teacher role as Read Only, and one shares Submission Details from the Teacher role to the Teacher role as Read/Write. Together with the profiles they are intended to let teachers read the assignments and work with the submissions. The restrictions in the right-hand column of Table 4.7 are design intent taken from the profile descriptions; no sign-in test with a restricted account was captured (Section 6.4).

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application was created in a Zoho Creator account under the name Student Assignment Management System. Its menu carries the initials “STU” on the application icon. The components are organised in the application menu as four sections and a dashboard page (Figure 4.6). The sections hold the four forms of the application (Student Details, Subject Details, Assignment Details and Submission Details) together with their reports, which are listed in Table 5.3. Five form workflows are attached to the forms: four on Submission Details and one on Assignment Details (Section 5.6).

The application was developed in the Development environment (the workflow screen offers the button “Access Development Live”), and version 2.0 was published to Production on 1 November 2026 at 15:14 IST. Figure 5.1 shows the notification e-mail that Zoho Creator sent after the publish. Table 5.1 summarises the set-up.

Table 5.1: Application Set-up Summary

Attribute	Details
Application name	Student Assignment Management System
Application icon	“STU”
Edition	Trial edition (about 15 days remaining when the screenshots were taken)
Menu	Student Details, Student Details, Assignment Details, Submission Details, dashboard
Environments	Development and Production
Version published	2.0, on 1 November 2026 at 15:14 IST
Users	One active user, the account owner (Super Admin)

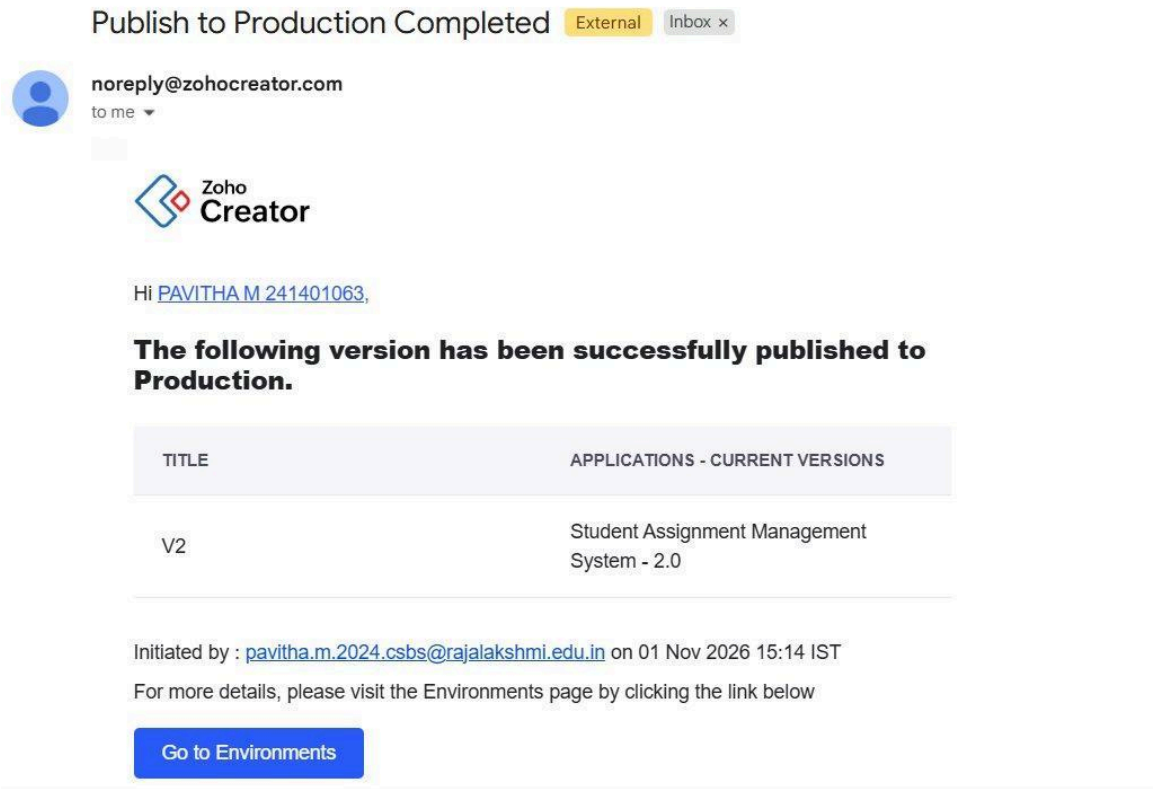


Figure 5.1: Zoho Creator Notification "Publish to Production Completed" for Version 2.0

5.2 Forms and Data Capture

The application stores its data in four forms: Student Details, Subject Details, Assignment Details and Submission Details. Table 5.2 summarises the forms; the full field specification is given in Tables 4.2 to 4.5 of Chapter 4.

Table 5.2: Forms Implemented

Form	Sections	Mandatory Fields	Special Features
Student Details	Single page: Student ID, Student Name, Email Address, Department, Phone number	Department	Phone field with country-code selector; drop-down for Department
Subject Details	Single page: Subject ID, Subject Name, Faculty Name, Semester	None marked	Drop-down for Semester
Assignment Details	Assignment Title, Subject, Description, Assigned Date, Due Date, Maximum Marks	Not shown	Lookup to Subject Details; 1 workflow (Calendar); form not captured in a screenshot
Submission Details	Student, Assignment, Submission Date, Submission Status, Marks Obtained, Feedback	Checked by the validation script	Lookups to Student Details and Assignment Details; 4 workflows

Student Details form. Figure 5.2 shows the form in live mode. It holds Student ID, Student Name, Email Address, Department (marked mandatory with an asterisk, drop-down) and Phone number with a +91 country-code selector.

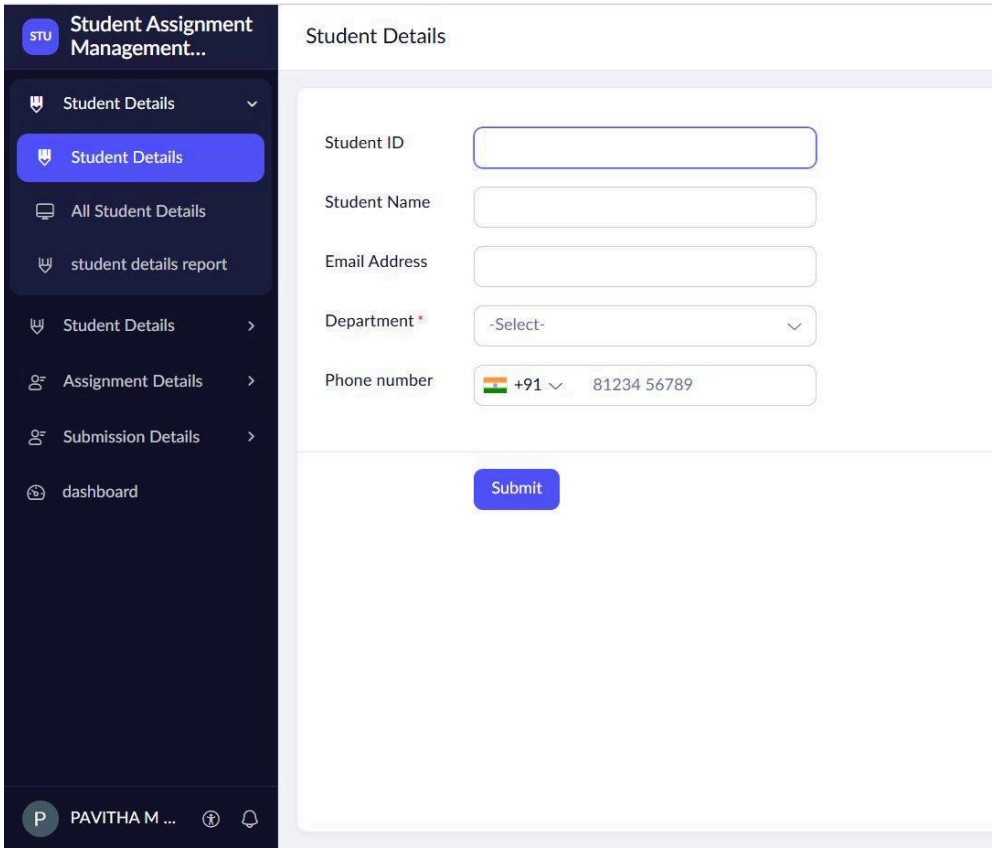


Figure 5.2: Student Details Form in Live Mode

Subject Details form. Figure 5.3 shows the form. It holds Subject ID, Subject Name, Faculty Name and a Semester drop-down.

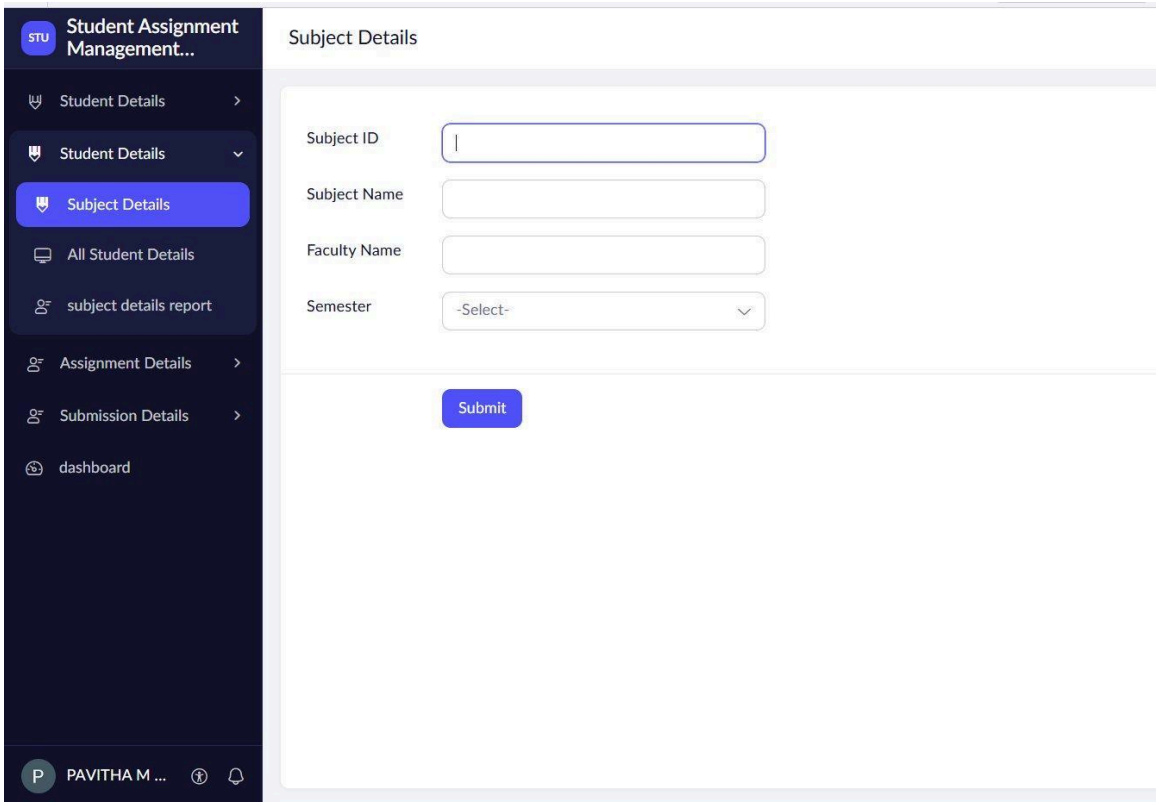


Figure 5.3: Subject Details Form in Live Mode

Submission Details form. Figure 5.4 shows the form in the builder and Figure 5.5 shows it in live mode. The builder lists the fields Student, Assignment, Submission Date, Submission Status, Marks Obtained and Feedback. Student and Assignment are lookup fields; the Field Properties panel for Student shows the display field Student Name from Student Details, the display type Drop Down and the Mandatory check box. In live mode the Student, Assignment and Submission Status fields are drop-downs, Submission Date shows the placeholder dd-MMM-yyyy, Marks Obtained is a number field and Feedback is a multi-line field. The Submission Details form therefore gives the many-to-one relationships shown in Figure 4.2. The Assignment Details form was not captured in a screenshot; its fields are taken from the Assignment Details Report (Figure 5.7).

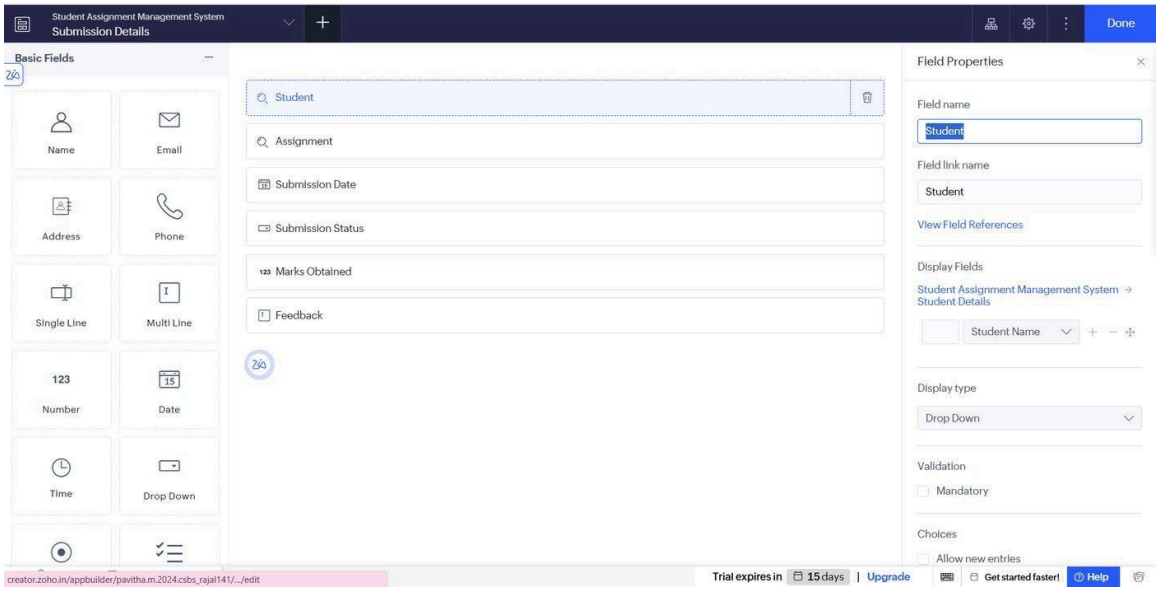


Figure 5.4: Submission Details Form in the Form Builder with the Student Lookup Properties

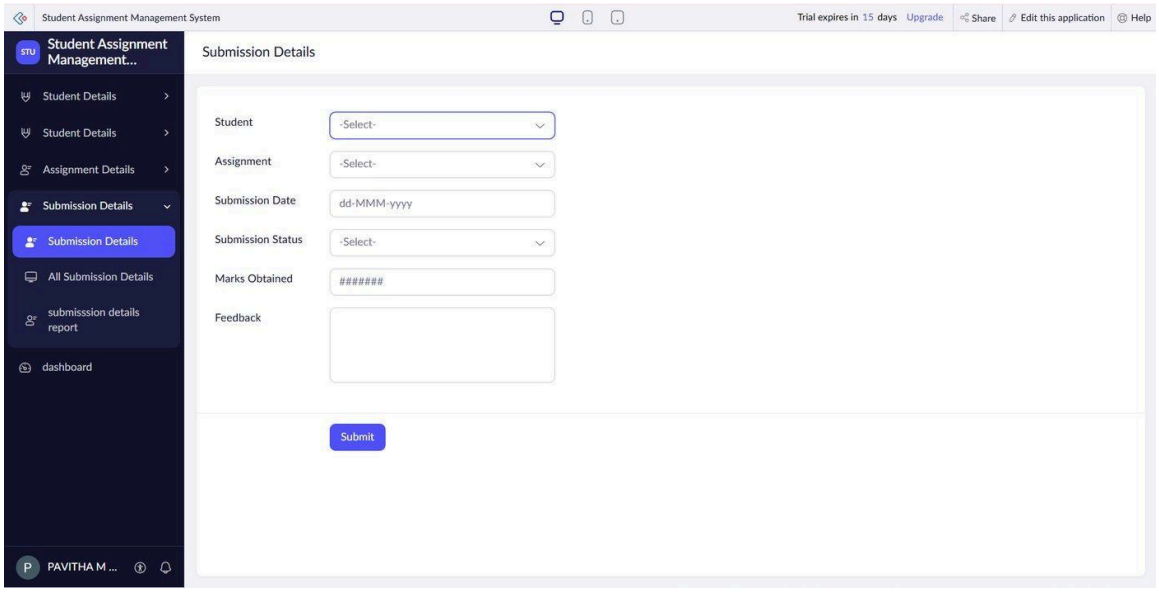


Figure 5.5: Submission Details Form in Live Mode

5.3 Reports

The application menu lists six report entries, which Table 5.3 summarises.

The student details report (Figure 5.6) is a list report of the Student Details form with the columns Student ID, Student Name, Email Address and Department, and it shows “Showing 7 of 7” records. The Assignment Details Report (Figure 5.7) lists Assignment Title, Subject, Description, Assigned Date, Due Date and Maximum Marks for four assignments (eessay, work, cnn and assignment1); the first has a blank description. The Submission Details reports and the subject report were not captured in screenshots, so their output is not shown.

Table 5.3: Reports Implemented

Report Name	Menu Section	Type	Source Form	Evidence / Purpose
All Student Details	Student Details	List	Student Details	Output not shown; lists the student records
student details report	Student Details	List	Student Details	Figure 5.6; seven records
subject details report	Student Details (second)	List	Subject Details	Output not shown
Assignment Details Report	Assignment Details	List	Assignment Details	Figure 5.7; four records
All Submission Details	Submission Details	List	Submission Details	Output not shown
submission details report	Submission Details	List	Submission Details	Output not shown

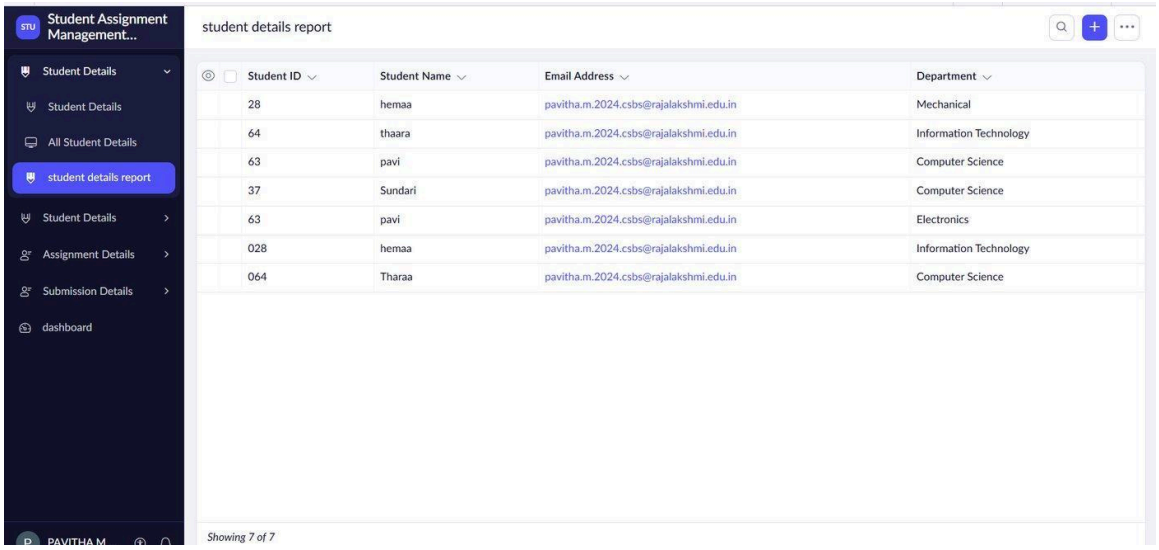


Figure 5.6: Student Details Report

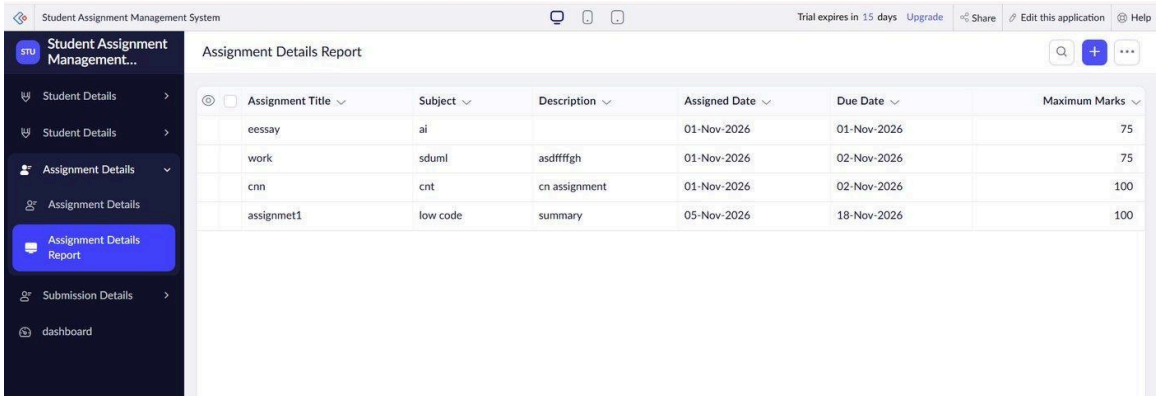


Figure 5.7: Assignment Details Report

5.4 Pages and Dashboards

The dashboard page (Figure 5.8) is the last entry in the menu and is built with the page builder. From top to bottom it contains a tile with the number of students, a pie chart, a tile with the number of subjects, two buttons ("submit assignment" and "Update submission status") and a

status panel titled “Order Status” with Submitted and Evaluated entries. Table 5.4 lists the components and the values visible in the screenshot.

The values do not all agree with the reports. The student tile shows 5 while the student details report shows 7 records (Figure 5.6), which is recorded as defect D-01 in Section 6.4. The pie chart legend shows the values Unknown, 50, 55 and 100, which suggests that the chart groups records by marks; its data source is not visible in the screenshot, so the chart is described only as it appears.

Table 5.4: Components of the Dashboard

Component	Type	Data Source	Value Observed
No of students	Count tile	Student Details	5
Pie chart	Chart	Not shown	Legend: Unknown, 50, 55, 100
No of subjects	Count tile	Subject Details	7
submit assignment	Button	Submission Details form	Shortcut button
Update submission status	Button	Submission Details report	Shortcut button
Order Status	Status panel	Not shown	Entries Submitted and Evaluated

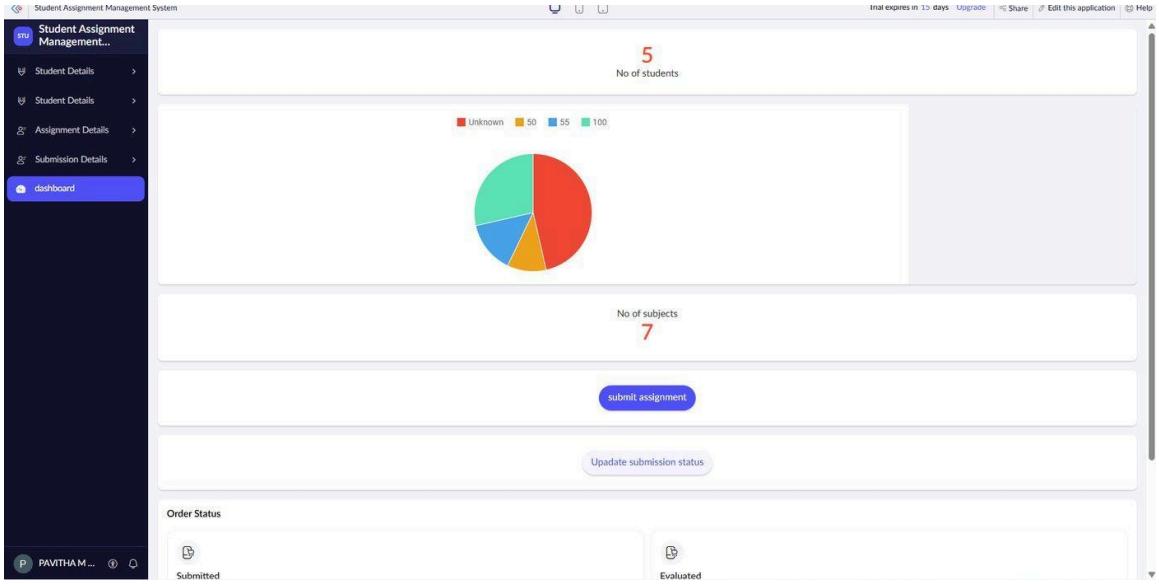


Figure 5.8: Dashboard Page

5.5 Deluge Scripting

Deluge is the scripting language of Zoho Creator; it is used to write form-event logic, functions and integrations, for example to create or update records [2]. All five form workflows run Deluge actions, and Table 5.5 lists them. Only two script bodies are visible in the supplied screenshots, Validate Submission Details and Check Late Submission; they are reproduced in Codes 5.1 and 5.2 and in Appendix C.

Table 5.5: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose
DS-01	Submission Details → Created or Edited → Validations on form submission (workflow "Validate Submission Details")	Alerts and cancels the submission if the assignment, student or status is empty (Code 5.1)
DS-02	Submission Details → Created or Edited → Validations on form submission (workflow "Check Late Submission")	Sets the submission status to Late or Submitted by comparing dates (Code 5.2)
DS-03	Submission Details → Validations on form submission (workflow "Student Status Update Notification")	E-mails the student the assignment ID and the current status (Figure 5.13); script not visible
DS-04	Submission Details → Successful form submission (workflow "Admin Submission Notification")	E-mails the administrator the student, assignment and status (Figure 5.12); script not visible
DS-05	Assignment Details → Successful form submission (workflow "Calendar")	Runs one action intended to create a Zoho Calendar event; script not visible

Validate Submission Details (DS-01). The script checks in turn whether the Assignment, Student and Submission Status fields are empty with the method isNull. For each empty field it shows an alert and cancels the submission with the statement cancel submit. Code 5.1 reproduces the script as it appears in the editor (Figure 6.1).

Code 5.1: Validate Submission Details (Submission Details → Created or Edited → Validations on form submission)

```
if(input.Assignment.isNull())
{
    alert "Please select an assignment.";
    cancel submit;
}
if(input.Student.isNull())
{
    alert "Please select a student.";
    cancel submit;
}
if(input.Submission_Status.isNull())
{
    alert "Please select a submission status.";
    cancel submit;
}
```

Check Late Submission (DS-02). When both the Assignment and the Submission Date are filled, the script fetches the Assignment Details record whose ID equals the selected assignment. If at least one record is found, it compares the submission date with the record's due date and sets the status variable to "Late" when the submission date is later, and to "Submitted" otherwise. Code 5.2 reproduces the script (Figure 6.2).

Code 5.2: Check Late Submission (Submission Details → Created or Edited → Validations on form submission)

```
if(!input.Assignment.isNull() && !input.Submission_Date.isNull())
{
    assignmentRecord = Assignment_Details[ID == input.Assignment];
    if(assignmentRecord.count() > 0)
    {
        if(input.Submission_Date > assignmentRecord.Due_Date)
        {
            submission_status = "Late";
        }
        else
        {
            submission_status = "Submitted";
        }
    }
}
```

5.6 Workflows and Automation

Five form workflows are defined, as listed in Table 5.6 and shown in Figure 5.9 and Figure 5.10. All are enabled, and the Student Details and Subject Details forms have no workflow.

Table 5.6: Workflows Implemented

Workflow Name	Form	Event / Trigger	Status	Purpose
Validate Submission Details	Submission Details	Created or Edited → Validations on form submission	Enabled (1 action)	Blocks incomplete submissions (DS-01)
Check Late Submission	Submission Details	Created or Edited → Validations on form submission	Enabled (1 action)	Sets Late or Submitted (DS-02)
Student Status Update Notification	Submission Details	Created or Edited → Validations on form submission	Enabled (1 action)	E-mails the student the status (DS-03)
Admin Submission Notification	Submission Details	Created or Edited → Successful form submission	Enabled (1 action)	E-mails the administrator (DS-04)
Calendar	Assignment Details	Created → Successful form submission	Enabled (1 action)	Creates a calendar event (DS-05)

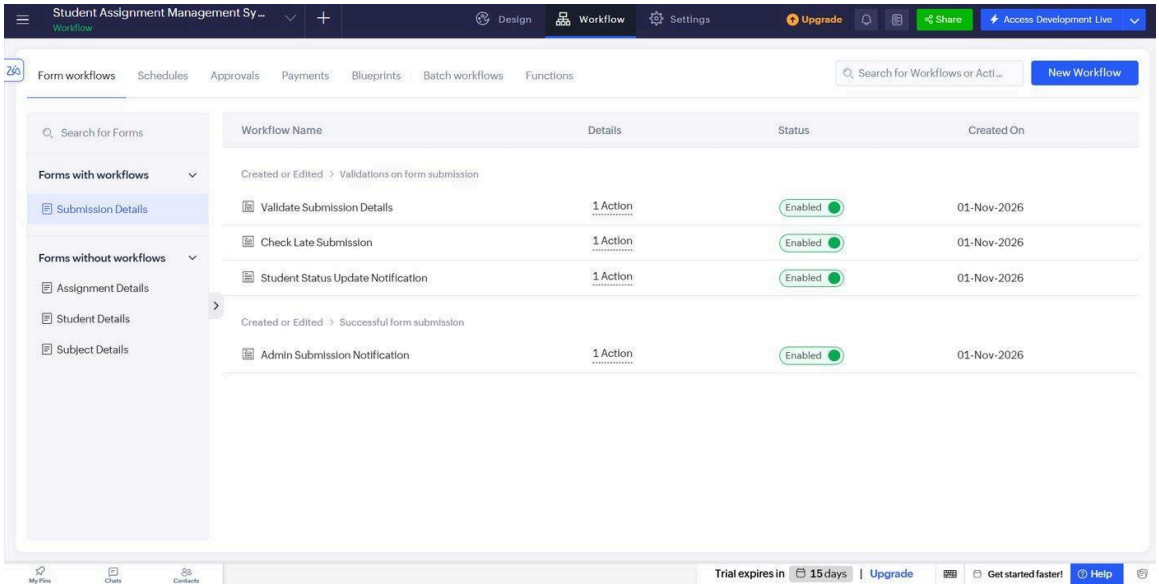


Figure 5.9: Form Workflows of the Submission Details Form

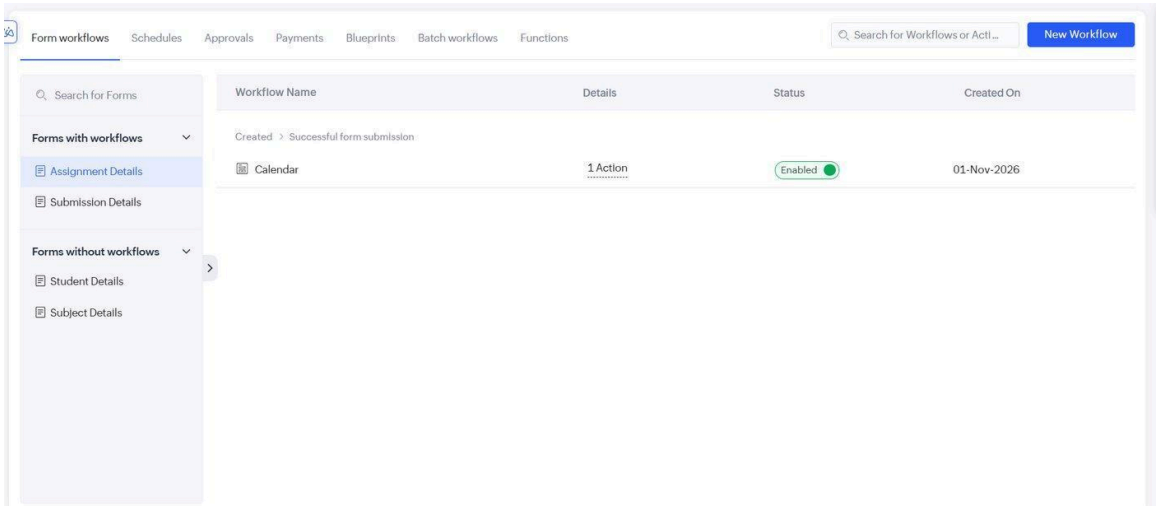


Figure 5.10: Form Workflow of the Assignment Details Form

Figure 5.11 shows the References dialog, which lists the Calendar workflow of the form Assignment Details in the Development environment. No schedule, approval, blueprint, payment or batch workflow is evidenced in the supplied screenshots, so automatic reminders before the due date are not implemented. One point is recorded for follow-up: the Student Status Update Notification workflow is listed under “Validations on form submission”, which runs before the record is saved, whereas a notification is normally sent after a successful save (D-13).

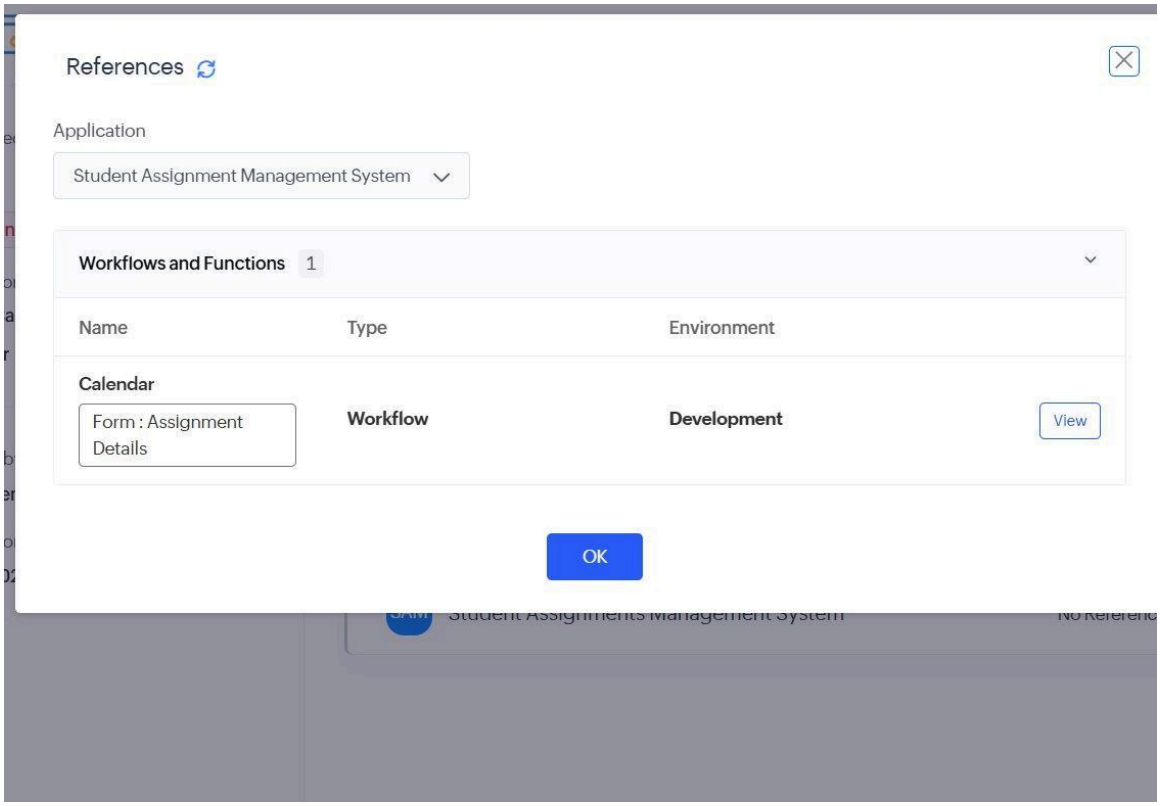


Figure 5.11: References Dialog Showing the Calendar Workflow of Assignment Details

5.7 Integrations

Table 5.7 summarises the integrations. **E-mail.** Two e-mails were received after submissions were recorded. The first has the subject "New Assignment Submission" and the text "A student submission has been recorded." followed by the student, assignment and status; it is addressed to the administrator (Figure 5.12). Two copies are visible: one for the student Tharaa, assignment eessay and status Submitted, and one for the student pavi, assignment eessay and status Evaluated. The second has the subject "Assignment Submission Status Updated"; it tells the student that the submission status has been updated and states the assignment ID 498310000000012034 and the current status Evaluated (Figure 5.13). Both come from the Zoho Creator trial mail address notifications@trial.zohocreatormail.com. **Zoho Calendar.** The Calendar workflow of Assignment Details is configured to create a calendar event for an assignment through a Zoho Calendar connection. The events themselves were not captured in a screenshot, so this integration is recorded as configured but not verified.

Table 5.7: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Administrator e-mail	Zoho Creator mail service	Mail action in Admin Submission Notification	Creator system sender	Student, assignment and status of a new submission
Student e-mail	Zoho Creator mail service	Mail action in Student Status Update Notification	Creator system sender	Assignment ID and current status

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Calendar event	Zoho Calendar	Calendar action in the Calendar workflow (one action)	Zoho Calendar connection	Assignment title and due date (intended); not verified

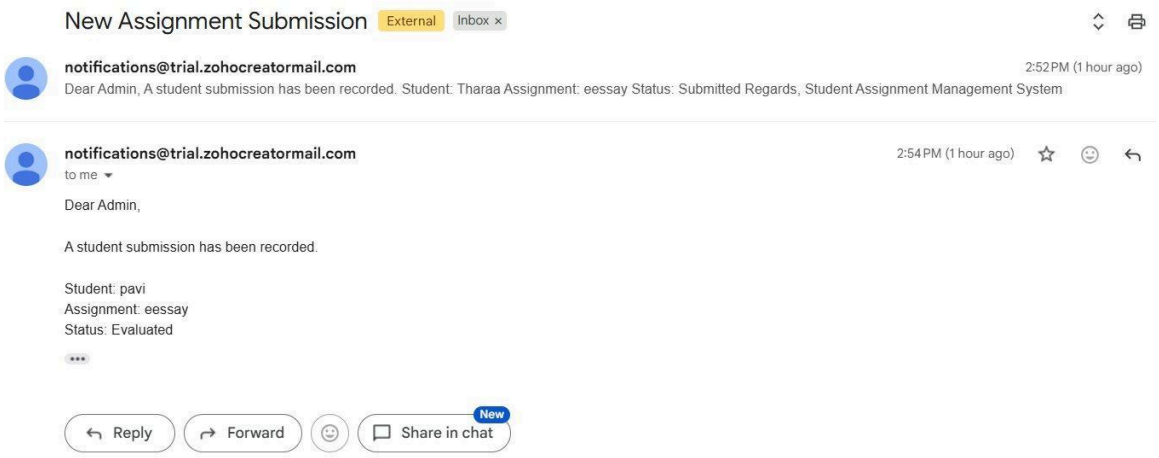


Figure 5.12: Administrator E-mail "New Assignment Submission"

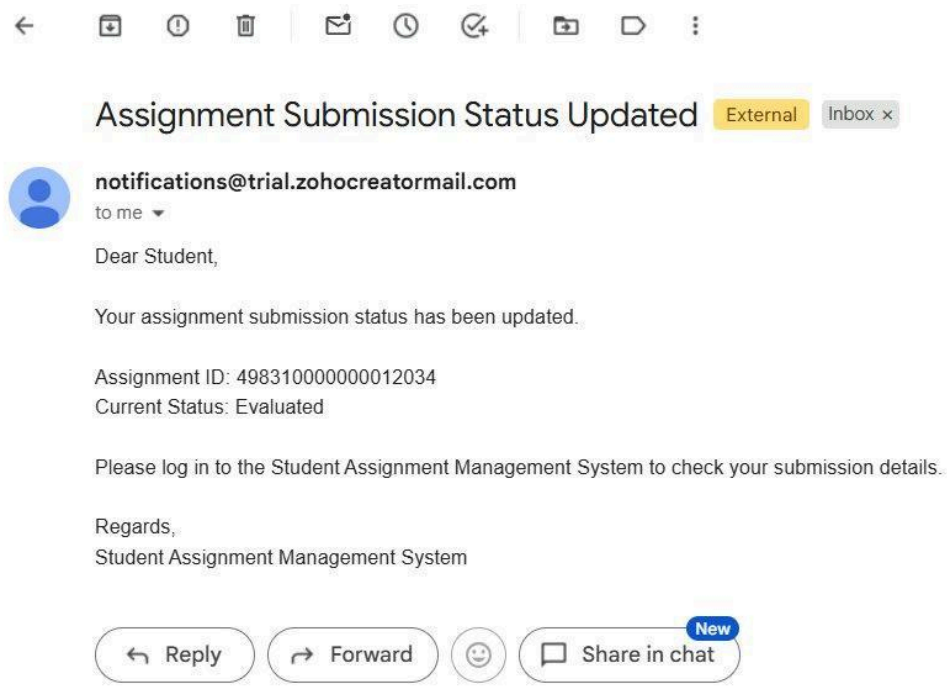


Figure 5.13: Student E-mail "Assignment Submission Status Updated"

5.8 User Management and Role-Based Access

Access control is built from roles, permission profiles and data sharing rules, all configured under Settings → User Permissions. Figure 5.14 shows the Roles tab. The hierarchy has CEO at the top, Admin and Teacher beneath it, and Student beneath Teacher. Figure 5.15 shows the permission profiles Admin_Permission, Read, Student_Permission, Teacher_Permission and Write; the descriptions of the first, third and fourth agree with the design intent recorded in Table 4.7.

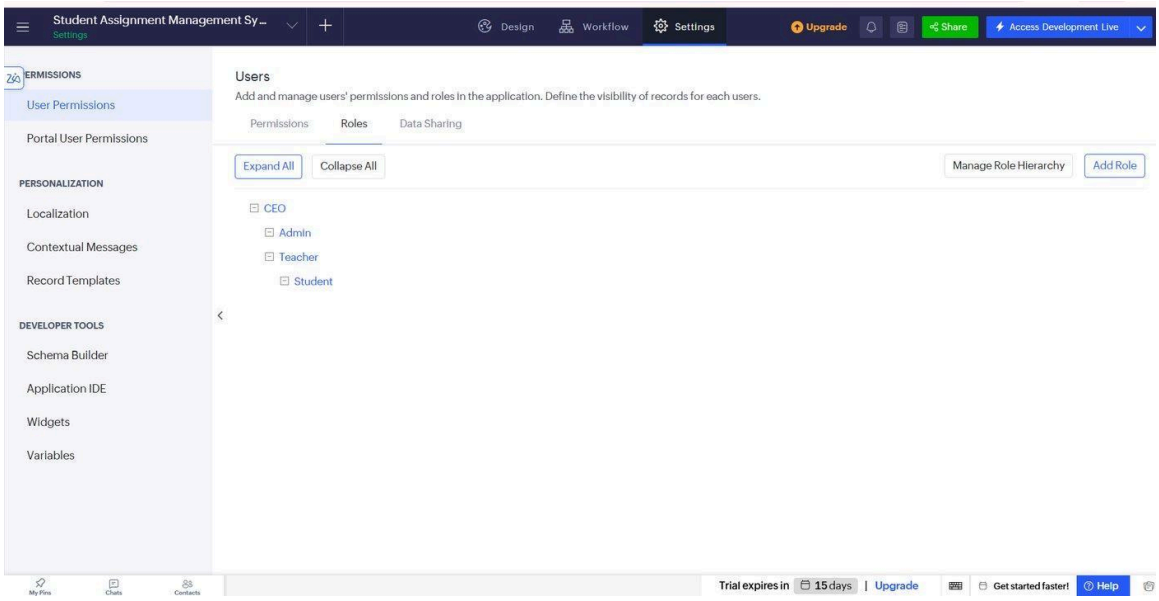


Figure 5.14: Role Hierarchy (CEO, Admin, Teacher and Student)

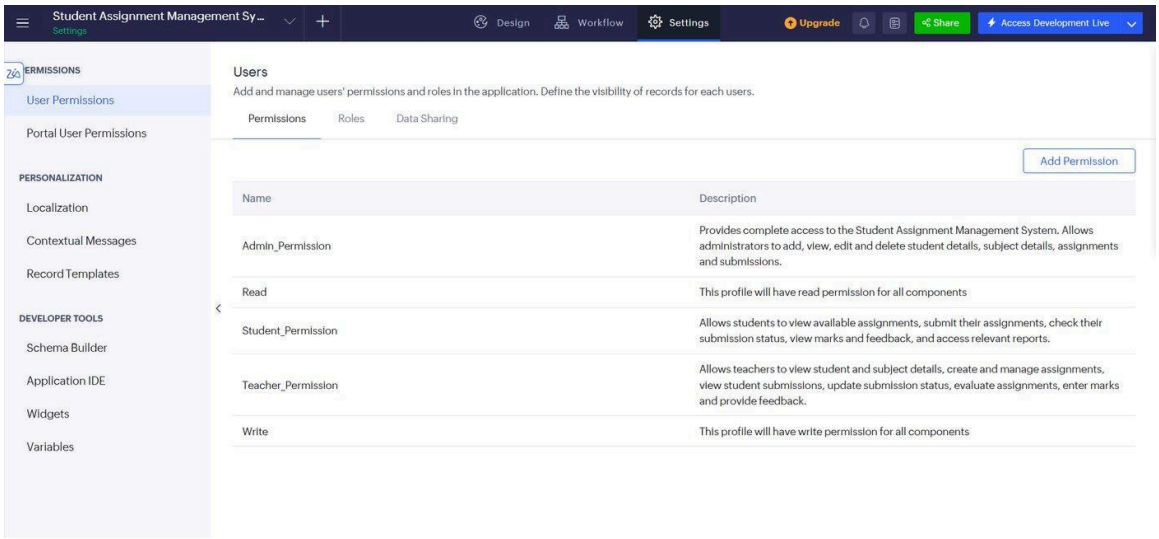


Figure 5.15: Permission Profiles of the Application

Figure 5.16 shows the Data Sharing tab, which holds two rules, summarised in Table 5.8. The rule “student” shares Assignment Details from the Student role to the Teacher role as Read Only.

The rule “faculty” shares Submission Details from the Teacher role to the Teacher role as Read/Write; because source and target are the same role, this rule may be redundant (D-15).

Table 5.8: Data Sharing Rules

Rule	Form	Shared From	Shared To	Level of Access
student	Assignment Details	R:Student	R:Teacher	Read Only
faculty	Submission Details	R:Teacher	R:Teacher	Read/Write

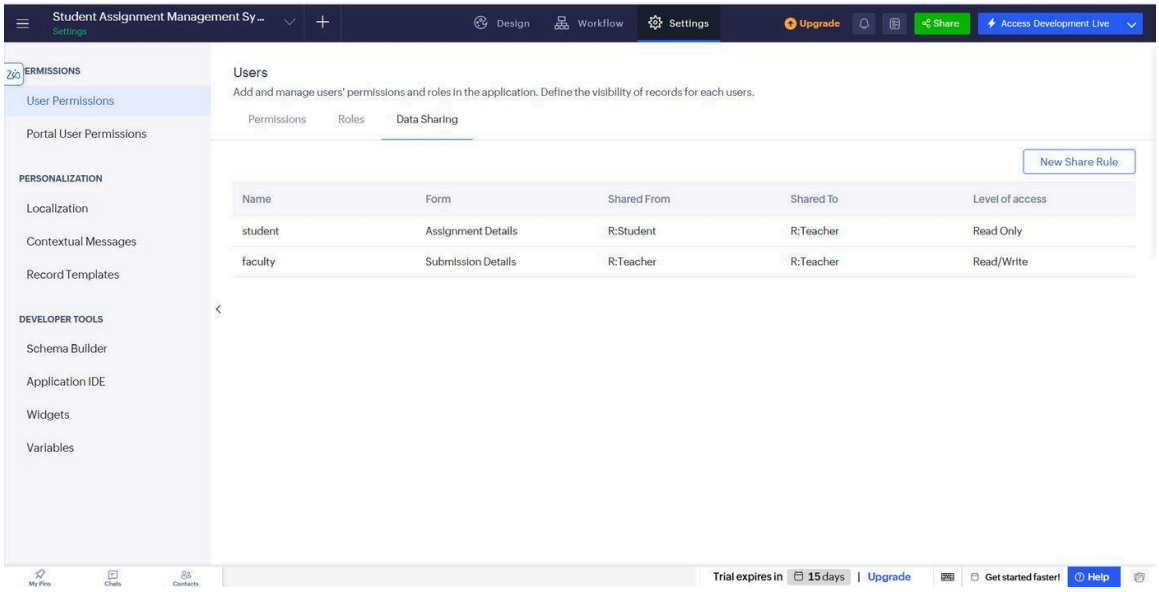


Figure 5.16: Data Sharing Rules

Figure 5.17 shows the users of the account: one active user, the account owner, with the label Super Admin. The assignment of users to the roles and profiles, a login test of each profile and the module-level permissions (access, view, edit, delete) are not shown, so the restrictions of the Teacher and Student profiles are recorded as configured but not tested (D-10).

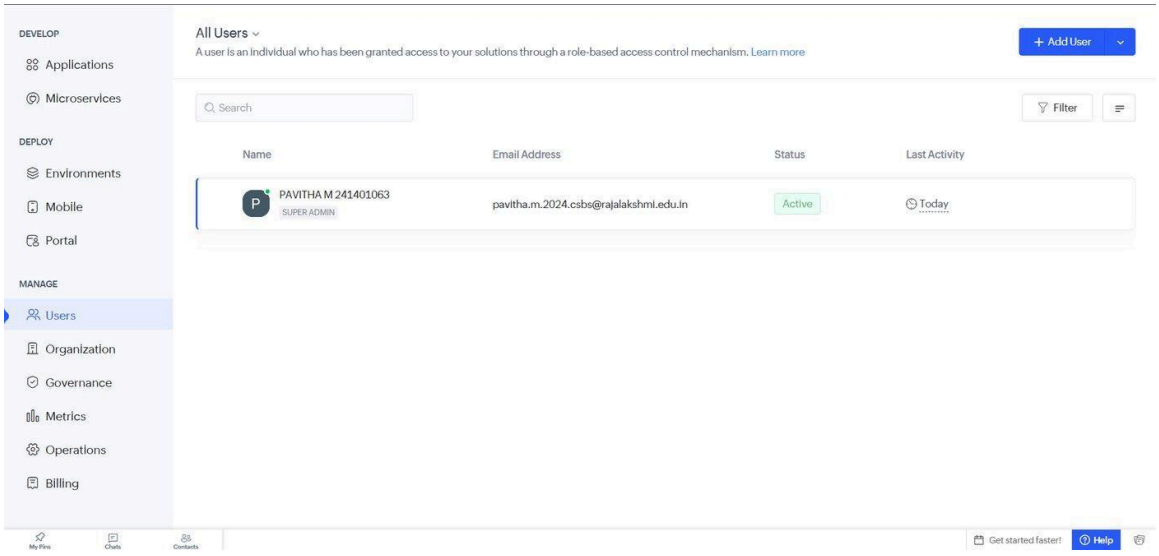


Figure 5.17: Users of the Account

5.9 AI and Analytics Features

No generative or predictive AI feature is evidenced in this application. The Zia assistant icon is visible in the builder screens, but no Zia-generated component or AI prediction is shown. Analytics are provided by the dashboard (Figure 5.8): two count tiles and a pie chart give a summary of the data. No Zoho Analytics integration is evidenced. Further charts, such as submissions per assignment and late submissions per subject, are listed as future enhancements in Section 9.2.

5.10 Pro-Code Extension

No pro-code extension was used. The application does not use Zoho Catalyst, custom widgets or an external service written in another language. All logic is implemented with Creator workflows and Deluge, which was sufficient for the validation, late-submission and notification requirements. A pro-code extension would be justified only if the project later needed heavy processing, such as plagiarism checking of uploaded files.

5.11 Mobile and Portal Access

Portal. The application is reached at the portal address https://pavitham2024csbsrajal141.zohocreatorportal.in. Figure 5.18 shows the sign-in page in the page designer: it asks for an e-mail address and offers Next, Forgot Password and Sign Up. A live portal sign-in by a student was not captured, so portal access is recorded as designed but not tested.

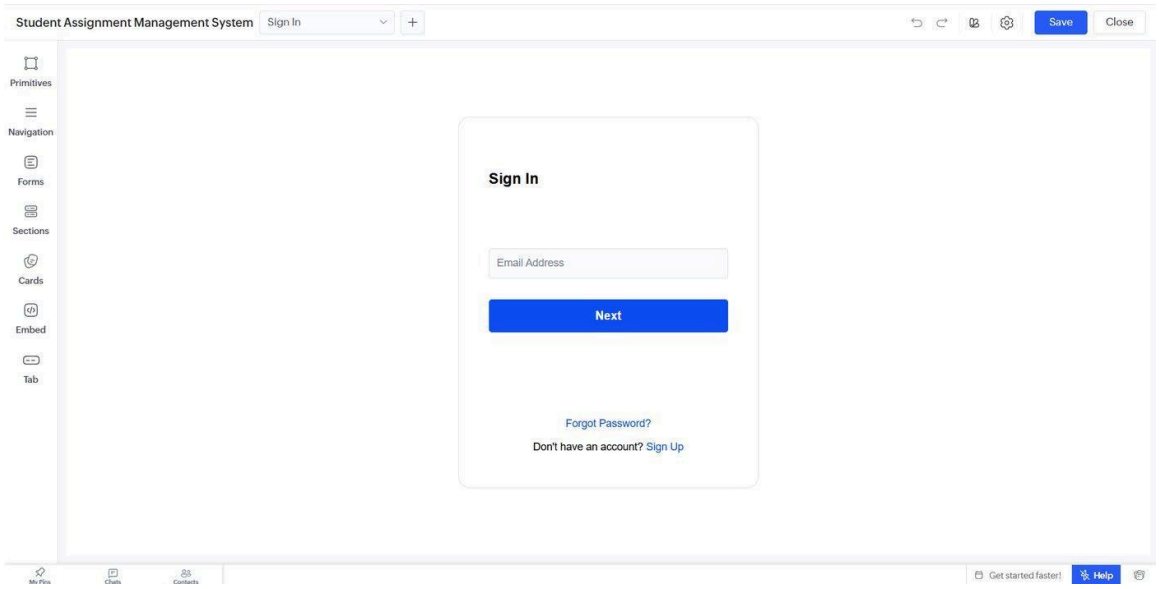


Figure 5.18: Sign-In Page of the Portal in the Page Designer

Mobile access. The application header offers desktop, tablet and phone preview icons (for example in Figure 5.5), and the Creator console lists a Mobile entry. The platform renders applications responsively, but no screenshot of the application on a phone or in the Zoho Creator mobile app was supplied.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was functional and manual. The application was exercised through its live interface in the Development environment of Zoho Creator, and version 2.0 was then published to Production on 1 November 2026 (Figure 5.1). Each feature described in Chapters 4 and 5 was checked by entering records, observing the reports and the dashboard, reading the e-mails produced by the workflows, and opening the Deluge scripts in the editor. The evidence for every test is a screenshot of the running application, so the test cases describe what the screenshots show and not a separate formal test run. A test is marked Pass only where a screenshot shows the result; where a screenshot shows a configuration but not its outcome, the test is marked Partly passed or Pending.

Checking was done at four levels:

- **Unit level.** Single forms and fields (layout, mandatory flags, drop-downs) and single scripts (reading the code in the editor).
- **Integration level.** Workflow to e-mail, workflow to calendar, and the dashboard figures compared with the contents of the reports.
- **Security level.** Roles, permission profiles and data sharing rules compared with their intended design.
- **Release level.** Publication from Development to Production.

6.2 Test Cases

Table 6.1 lists the test cases. Status values are Pass (evidenced), Partly passed, Pending and Fail.

Table 6.1: Test Cases

TC ID	Feature	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC01	Students	Create and list student records	Fill the Student Details form and submit	Records are saved and listed	The student details report shows 7 records (Figure 5.6)	Pass (evidenced)
TC02	Students	Mandatory department	Submit the form with Department empty	A validation message is shown	Department carries the mandatory asterisk (Figure 5.2); a blocked submission was not captured	Partly passed
TC03	Subjects	Subject form	Open the Subject Details form	Four fields are shown	Subject ID, Subject Name, Faculty Name and Semester are shown (Figure 5.3)	Pass (evidenced)
TC04	Subjects	Subject report	Open the subject details report	Subject records are listed	Report output not supplied; the dashboard tile shows 7 subjects (Figure 5.8)	Partly passed
TC05	Assignments	Create and list assignments	Fill the Assignment Details form and submit	Assignments appear with subject, dates and marks	Four assignments are listed with subject, dates and maximum marks (Figure 5.7)	Pass (evidenced)

TC ID	Feature	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC06	Assignments	Subject lookup	Choose a subject on the assignment	The subject name is stored	Subjects ai, sduml, cnt and low code appear in the report (Figure 5.7)	Pass (evidenced)
TC07	Submissions	Lookups on the submission form	Open the Submission Details form	Student and Assignment are chosen from lists	Student and Assignment are lookup fields with drop-downs (Figures 5.4 and 5.5)	Pass (evidenced)
TC08	Submissions	Empty-field validation	Submit with assignment, student or status empty	Alert shown and save cancelled	The validation script is present (Figure 6.1); a blocked submission was not captured	Partly passed
TC09	Submissions	Late-submission detection	Submit with a date after the due date	Status becomes Late	The script is present (Figure 6.2); the admin e-mail shows the status Submitted for one record (Figure 5.12); a Late case was not captured	Partly passed
TC10	Notifications	Administrator e-mail	Record a submission	The administrator receives the e-mail	“New Assignment Submission” e-mails received with student, assignment and status (Figure 5.12)	Pass (evidenced)
TC11	Notifications	Student e-mail	Update the submission status	The student receives the e-mail	“Assignment Submission Status Updated” received with status Evaluated (Figure 5.13)	Pass (evidenced)
TC12	Integration	Calendar event	Create an assignment	An event appears in Zoho Calendar	The Calendar workflow is enabled (Figure 5.10); no event was captured	Pending
TC13	Workflows	Workflow status	Open the workflow list	All workflows are enabled	Five workflows show the status Enabled (Figures 5.9 and 5.10)	Pass (evidenced)
TC14	Dashboard	Counts agree with reports	Compare the student tile with the student report	Both show the same count	The tile shows 5 and the report shows 7 (Figures 5.6 and 5.8)	Fail
TC15	Dashboard	Shortcut buttons	Click “submit assignment” and “Update submission status”	The forms and reports open	The buttons are shown (Figure 5.8); the click-through was not captured	Pending
TC16	Dashboard	Chart readability	Read the pie chart	Segments are labelled by a meaningful value	The legend shows Unknown, 50, 55 and 100 (Figure 5.8)	Partly passed
TC17	Security	Role hierarchy	Open Settings → Roles	CEO, Admin, Teacher and Student are listed	The hierarchy is shown (Figure 5.14)	Pass (evidenced)
TC18	Security	Permission profiles	Open Settings → Permissions	The three profiles are listed with descriptions	Admin_Permission, Student_Permission and Teacher_Permission are shown (Figure 5.15)	Pass (evidenced)
TC19	Security	Data sharing rules	Open the Data Sharing tab	Two rules are listed	The rules “student” and “faculty” are shown (Figure 5.16)	Pass (evidenced)
TC20	Security	Role-based access with test users	Sign in as Teacher and as Student and open each form	Each role sees only the permitted data	Only one user exists (Figure 5.17), so no test was possible	Pending
TC21	Portal	Sign-in page	Open the portal sign-in page	Sign-in page loads	The sign-in page is designed (Figure 5.18); a live sign-in was not captured	Partly passed
TC22	Data quality	Unique student ID	Add a second student with an existing ID	The duplicate is rejected	Student ID 63 appears twice in the report (Figure 5.6)	Fail
TC23	Navigation	Menu names	Read the side menu	Each section has a distinct, clear name	Two sections are both named “Student Details” (Figure 5.2)	Fail
TC24	Release	Publish to Production	Publish version 2.0	Publication is confirmed	Zoho Creator confirms version 2.0 published on 1 Nov 2026 at 15:14 IST (Figure 5.1)	Pass (evidenced)

Table 6.2: Summary of Test Results

Status	Count	Test Cases
Pass (evidenced)	12	TC01, TC03, TC05, TC06, TC07, TC10, TC11, TC13, TC17, TC18, TC19, TC24
Partly passed	6	TC02, TC04, TC08, TC09, TC16, TC21
Pending	3	TC12, TC15, TC20
Fail	3	TC14, TC22, TC23
Total	24	

No test case failed because of a script error; the three failures are design and data gaps (D-01, D-02 and D-03 in Section 6.4 give the detail). The pending tests are pending because the screenshots show a configuration but not its outcome, and they close as soon as the missing screenshot or sign-in test is captured. The screenshots cited in Table 6.1 are shown once, in Chapter 5, and are not repeated here.

6.3 Debugging Deluge Scripts

Three debugging aids are available in the Deluge editor: the Execute button, the info statement that writes a value to the log, and the output that each workflow leaves behind (an e-mail or a calendar entry). Two script bodies are visible in the editor, and the workflow list makes it possible to trace which script produced which output.

Validate Submission Details. This workflow runs when a Submission Details record is created or edited and saved, before the record is stored (Figure 5.9). The script is listed in Code 5.1 and shown in the editor in Figure 6.1. It tests three fields with isNull and cancels the submission with an alert for each empty field. Reading the code gives two points to test. First, the script does not test the Submission Date, so a record can be saved without one. Second, the three tests are separate if statements, so the user sees one alert at a time.

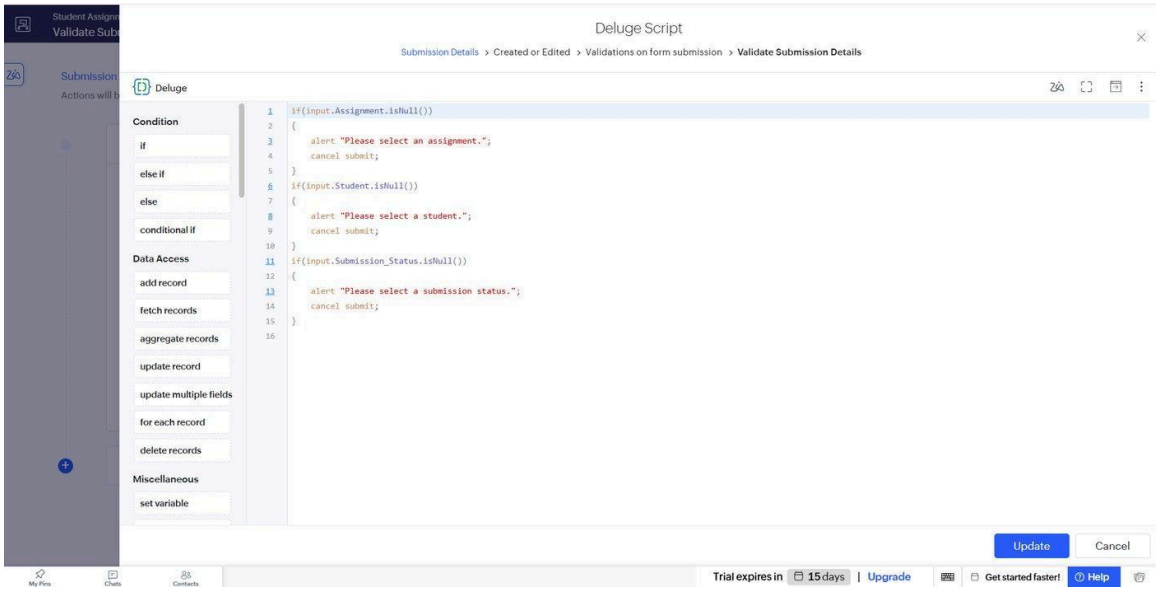


Figure 6.1: Deluge Script of the Validate Submission Details Workflow

Check Late Submission. This workflow runs on the same event (Figure 5.9). The script is listed in Code 5.2 and shown in the editor in Figure 6.2. If the assignment and the submission date are both filled, it fetches the assignment record by its ID and compares the dates. Reading the code gives three points to test. First, nothing is set when the date is empty, so the status stays as chosen by hand (D-14). Second, the comparison is on dates only, so a submission on the due date is Submitted. Third, the script assigns the status variable but does not show a message, so its effect can be seen only in the saved record.

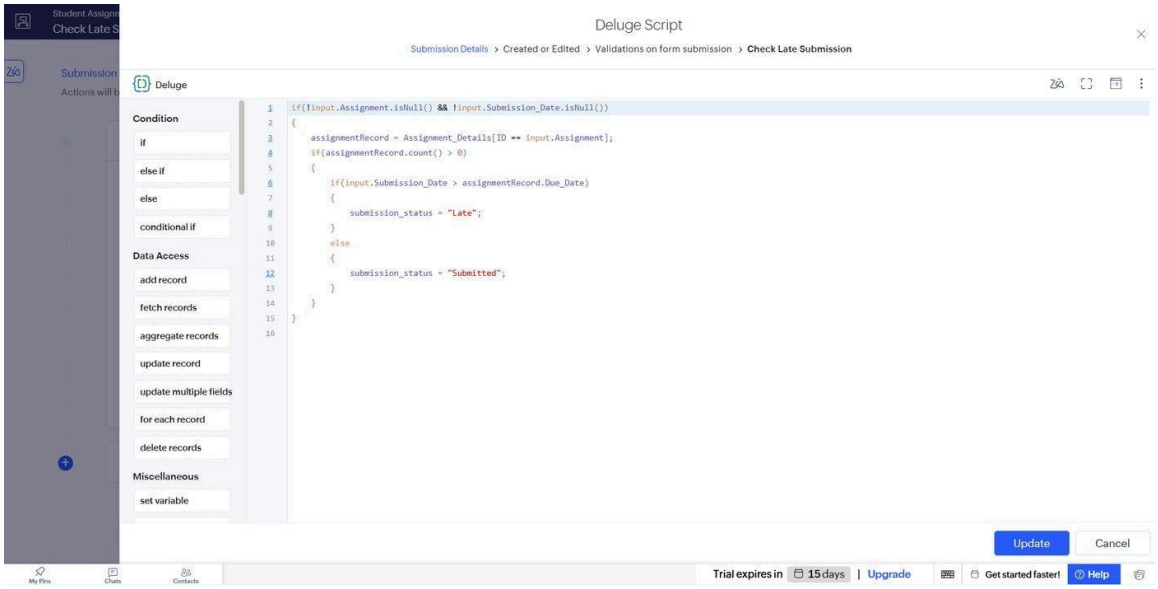


Figure 6.2: Deluge Script of the Check Late Submission Workflow

Table 6.3 lists the symptoms that were traced in this way and what each trace showed.

Table 6.3: Debugging Observations

No.	Symptom	Technique Used	Finding	Defect
1	The dashboard shows 5 students but the report shows 7	Compared the tile with the student report	The tile and the report count differently; the criteria of the tile are not visible	D-01
2	Student ID 63 appears twice	Read the student report	There is no unique constraint on Student ID	D-02
3	The student e-mail shows a long assignment ID instead of a title	Read the e-mail	The message prints the assignment lookup value, which is a record ID	D-08
4	The student notification sits under “Validations on form submission”	Read the workflow list	The event runs before the record is saved, so the e-mail may be sent for a submission that is later cancelled	D-13
5	A submission saved without a date has no automatic status	Read the late-submission script	The condition requires a non-empty date	D-14

6.4 Defect Log

Table 6.4 records the issues found while preparing the evidence for this chapter. D-01 to D-03 are the failed test cases; the others are observations. All items are open, and the proposed fixes have not been deployed or tested. Root causes marked “Suspected” were inferred from the screenshots and were not confirmed in the builder.

Table 6.4: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
D-01	The dashboard tile shows 5 students while the student report shows 7 records (Figures 5.6 and 5.8).	Suspected: the tile counts with different criteria or a stale value.	Proposed: recreate the count expression on the Student Details form.	Open
D-02	Student ID 63 appears on two records (Figure 5.6).	No unique constraint on Student ID.	Proposed: mark Student ID as unique.	Open
D-03	The menu has two sections named “Student Details”, and “All Student Details” appears under both (Figure 5.2).	Menu items were named by default or copied during configuration.	Proposed: rename the second section to Subject Details.	Open
D-04	One report is named “submisssion details report” with a spelling error and lower-case names (Figure 5.5).	Names typed by hand.	Proposed: rename the reports in a consistent style.	Open
D-05	The dashboard still has a panel titled “Order Status” (Figure 5.8).	A default label was not changed.	Proposed: rename it to Submission Status.	Open
D-06	The pie chart legend shows Unknown, 50, 55 and 100, which does not say what is measured (Figure 5.8).	Suspected: the chart groups by Marks Obtained and some records have no marks.	Proposed: group by Submission Status and add a title.	Open
D-07	The administrator e-mail is also sent when the status is Evaluated (Figure 5.12).	The workflow runs on every successful save.	Proposed: decide whether the message should be sent only for new submissions.	Open

Bug ID	Description	Root Cause	Fix Applied	Status
D-08	The student e-mail shows the assignment ID 498310000000012034 instead of the assignment title (Figure 5.13).	The lookup value is printed as a record ID.	Proposed: print the assignment title by fetching the record.	Open
D-09	E-mails come from the trial mail address notifications@trial.zohocreatormail.com.	The account is in trial.	Proposed: configure the organisation's sender address after upgrade.	Open
D-10	Only one user exists, so the restrictions of the Teacher and Student profiles were not tested (Figure 5.17).	No test users were created.	Proposed: create one user per role and test each.	Open
D-11	No screenshot shows the calendar event created by the Calendar workflow.	Test not captured.	Proposed: create an assignment and capture the event.	Open
D-12	No run of the validation and late-submission scripts was captured.	Tests not captured.	Proposed: record one blocked submission and one Late case.	Open
D-13	The Student Status Update Notification workflow runs under "Validations on form submission" (Figure 5.9).	Suspected: the workflow was created under the wrong event.	Proposed: move it to Successful form submission.	Open
D-14	Check Late Submission sets nothing when the Submission Date is empty, and Validate Submission Details does not require the date.	The script condition requires a date.	Proposed: add a date check to the validation script.	Open
D-15	The data sharing rule "faculty" shares Submission Details from Teacher to Teacher (Figure 5.16).	Suspected: the source and target roles were set to the same role.	Proposed: review the rule and delete it if it is redundant.	Open
D-16	Sample data contain test values such as the title "eessay" and a meaningless description (Figure 5.7).	Test data entered during development.	Proposed: remove or correct the sample records before go-live.	Open

6.5 User Acceptance Testing

No formal user acceptance session was recorded for this version, so no user feedback is reported. Table 6.5 is kept as the form in which the feedback of teachers and students will be recorded once the application is trialled.

Table 6.5: User Acceptance Feedback

Reviewer	Role	Task Tried	Feedback	Action Taken
Not recorded	–	–	No user acceptance session was recorded for version 2.0	–

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator separates an application into Development, Stage and Production environments: changes are made and tried in Development and then published forward, so that real users work on a stable copy. The Environments entry is part of the console navigation (Figure 5.17). The application is built in an application named "Student Assignment Management System". After development, version 2.0 of the application was published to Production on 1 November 2026 at 15:14 IST (Figure 5.1).

The deployment followed the steps below. Steps 1 to 3 are supported by the evidence; steps 4 and 5 are listed for completeness.

1. Develop and unit-test forms, reports, the dashboard and workflows in the Development environment (the workflow screen offers "Access Development Live").
2. Configure roles, permission profiles and data sharing rules (Section 7.3).
3. Publish version 2.0 to Production (Figure 5.1).
4. Share the portal URL with teachers and students and create their accounts.
5. Record release notes for each later version and repeat the publish from Development.

Table 7.1: Deployment Checklist

Step	Status	Evidence
Forms, reports, dashboard and workflows built in Development	Done	Chapter 5
Roles, profiles and data sharing rules configured	Done	Figures 5.14 to 5.16
Version 2.0 published to Production	Done, 1 November 2026, 15:14 IST	Figure 5.1
Tested with Admin, Teacher and Student accounts in Production	Pending	D-10
Sample data removed before go-live	Pending	D-16
Trial edition upgraded or application moved to a paid plan	Pending	About 15 days remained when the screenshots were taken

7.2 Performance Optimisation

No performance measurement was recorded, so no before-and-after figures are reported. The review of the scripts identified the opportunities in Table 7.2. They are proposals and have not been applied. The Check Late Submission script already fetches only the one assignment record it needs by ID, instead of fetching all assignments. The remaining opportunities concern when the scripts run and how much data the reports load.

Table 7.2: Performance Optimisation Opportunities (Proposed, Not Measured)

Proposed Optimisation	Component	Current Approach	Proposed Approach	Expected Benefit
Run the late check only when the date or assignment changes	Check Late Submission	Runs on every create or edit of a submission	Add a condition so that the script runs only when the assignment or the date changed	Fewer unnecessary fetches
Filter reports by default	Submission reports	List reports show all records	Add a default filter such as the current semester or the current assignment	Smaller pages as data grows
Move the student notification after the save	Student Status Update Notification	Runs under Validations on form submission	Run it on Successful form submission	Avoids e-mails for submissions that are cancelled
Remove unused menu entries	Menu	Duplicate section and report names	Rename or delete the duplicates	Easier navigation

7.3 Security Features

Table 7.3 lists the security controls and the evidence for each.

Table 7.3: Security Controls

Control	Implementation	Evidence / Gap
Authentication	Users sign in with an e-mail address; the portal sign-in page offers Forgot Password and Sign Up	Figure 5.18; live sign-in not captured
Role hierarchy	CEO, Admin, Teacher, Student	Figure 5.14
Permission profiles	Admin_Permission, Teacher_Permission, Student_Permission	Figure 5.15; module-level settings not shown
Record-level sharing	Two data sharing rules	Figure 5.16; one rule may be redundant (D-15)
Input validation	Submission blocked when assignment, student or status is empty	Figure 6.1; run not captured
Personal data	The Student Details form holds names, e-mail addresses and phone numbers	Data privacy options of the fields were not shown as selected

Data sharing. The two rules in Table 5.8 (Figure 5.16) let teachers read the assignments of the Student role and read and write submissions. Because the second rule has the same source and target role, it may be redundant. **Personal data.** The Student Details form collects names, e-mail addresses and phone numbers. The Field Properties panel of Zoho Creator offers Data Privacy and Data Security options; they were not shown as selected in the screenshots. Marking the personal fields and considering encryption is recommended before go-live. **Notification sender.** E-mails are sent from the Zoho Creator trial mail address; the sender should be changed to an organisation address after the trial.

7.4 Governance and Administration

Governance in Zoho Creator is exercised from the console, whose navigation (Figure 5.17) offers Users, Organization, Governance, Metrics, Operations and Billing. The Users page lists one user, who is Active and holds the Super Admin designation. The supplied evidence does not show the use of the audit trail, governance reports or metrics, so none is claimed.

The following practices are recommended for the production deployment:

- **Review.** Review the role hierarchy, the permission profiles and the data sharing rules at a fixed interval and remove any redundant rule.
- **Test users.** Create one user per role so that restrictions can be tested after each change.
- **Test data.** Remove sample students, subjects and assignments before go-live and keep a separate test subject for later trials.
- **Release notes.** Record each published version (the first is 2.0) with the workflows and scripts that changed.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

A single scenario illustrates the working application. An administrator enters students on the Student Details form and subjects on the Subject Details form (Figures 5.2 and 5.3). A teacher then creates an assignment on the Assignment Details form by choosing a subject and entering the dates and maximum marks; Figure 5.7 shows four such assignments, for example “assignment1” for the subject low code, assigned on 05-Nov-2026 and due on 18-Nov-2026 with a maximum of 100 marks. A submission is recorded on the Submission Details form by choosing the student and the assignment and entering the date and status (Figure 5.5). On saving, the validation and late-submission scripts run (Codes 5.1 and 5.2).

Two notifications follow a submission. First, an e-mail titled “New Assignment Submission” tells the administrator the student, assignment and status (Figure 5.12; for example the student Tharaa, the assignment eessay and the status Submitted). Second, when the teacher updates the status, an e-mail titled “Assignment Submission Status Updated” tells the student the assignment ID and the current status (Figure 5.13; here Evaluated). The records are listed in the reports (Figures 5.6 and 5.7), and the dashboard (Figure 5.8) shows the counts of students and subjects with a chart and shortcut buttons.

8.2 Achievement of Objectives

Table 8.1 maps the objectives of the project to the outcomes.

Table 8.1: Mapping of Objectives to Outcomes

Objective	Outcome	Status
Design linked Student, Subject, Assignment and Submission forms	Four forms were built; the assignment refers to a subject, and the submission to a student and an assignment (Chapters 4 and 5)	Achieved
Validate submissions and set Late or Submitted automatically	Both scripts are in place (Figures 6.1 and 6.2); a recorded run of each is pending (D-12)	Partly achieved
Notify the administrator and the student by e-mail	Both e-mails were received (Figures 5.12 and 5.13); the sender is the trial address and the student e-mail shows an ID (D-08, D-09)	Achieved; refinements open
Create a Zoho Calendar event for each assignment	The Calendar workflow is enabled (Figure 5.10); no event was captured (D-11)	Pending verification
Provide reports and a dashboard	List reports and a dashboard with tiles and a chart are in place (Figures 5.6 to 5.8); the student count disagrees with the report (D-01)	Achieved; one defect
Control access through roles, profiles and sharing rules	Four roles, three profiles and two rules are configured (Figures 5.14 to 5.16); no multi-user test (D-10)	Partly achieved
Publish from Development to Production	Version 2.0 published on 1 November 2026 (Figure 5.1)	Achieved

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Aspect	Existing (Manual) System	Student Assignment Management System
Records	Paper, e-mail and registers	Four linked forms in one application
Assignment details	Announced in class or on a messaging group	Assignment Details form and report with subject, dates and maximum marks
Late detection	By hand	Date comparison in a Deluge script
Status tracking	Mental note or tick in a register	Submission Status stored on each record (Submitted, Late, Evaluated)
Notification	None	E-mails to the administrator and the student
Marks and feedback	Separate register	Stored on the submission record
Summary view	Manual tally	Reports and a dashboard
Access control	Physical possession of the register	Roles, permission profiles and data sharing rules

8.4 Limitations of the Low-Code Approach

Several limitations were met in this project. First, platform behaviour that is easy to configure can be hard to inspect. The student e-mail that prints an assignment ID instead of a title (D-08) shows that event-driven logic depends on knowing exactly what a lookup field returns inside a script, which matches the finding that event handling is a difficult area of low-code development [5]. Second, the placing of a workflow on the wrong event, such as the notification under validations (D-13), is easy to do in a builder and is found only by reading the workflow list. Third, the dashboard counts depend on settings that are not visible on the page, so the mismatch of 5 and 7 (D-01) needed the reports to be compared. Fourth, the trial edition limits the time in which the application can be demonstrated, and the trial mail address is used as the sender. Finally, as practitioners also report, customisation beyond the built-in components is limited [4]; features such as file upload checks or plagiarism detection would need a pro-code extension.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The application sits mainly in the low-code range of the spectrum. Forms, lookups, reports, the dashboard, roles and publishing were done by configuration, which is the no-code part of the platform. The validation, the late-submission check and the notifications needed short Deluge scripts, which is the low-code part. No pro-code component was needed (Section 5.10). The project shows that a citizen developer with basic scripting skills can build a working academic workflow application, and that a professional developer would be needed for features that go beyond the platform, such as automatic reminders with custom rules, file analysis or integration with a college information system.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

Student Assignment Management System demonstrates a working low-code solution for managing assignments and submissions on Zoho Creator. The deployed application (Student Assignment Management System, version 2.0) provides four forms (Student Details, Subject Details, Assignment Details and Submission Details), reports for students, subjects, assignments and submissions, a dashboard with count tiles and a chart, five form workflows, e-mail notifications to the administrator and the student, a Zoho Calendar workflow, four roles, three permission profiles and two data sharing rules. A submission is validated and given the status Late or Submitted by a Deluge script, and the administrator and the student are informed by e-mail.

The review of the final screenshots also identified work that remains (Table 6.4): the dashboard student count disagrees with the report, Student ID is not unique, the menu has duplicate names, the student e-mail prints an ID, the calendar event and the role-based restrictions have not been verified, and the trial period is about to end. These items should be closed and re-tested before the application is relied upon in a live department.

9.2 Future Enhancements

- **Verification.** Capture the calendar event, a blocked submission, a Late submission and sign-in tests for the Teacher and Student roles.
- **Defect fixes.** Close the open items D-01 to D-16 of Table 6.4.
- **File upload.** Add a file-upload field so that students can attach their work to a submission.
- **Reminders.** Add a scheduled workflow that e-mails students before the due date.
- **Marks check.** Check that the marks obtained do not exceed the maximum marks of the assignment.
- **More analytics.** Add charts such as submissions per assignment and late submissions per subject.
- **Readable e-mails.** Show the assignment title and the due date in the student e-mail.
- **Privacy hardening.** Mark student name, phone and e-mail as personal data and consider field encryption.
- **Mobile access.** Configure the Zoho Creator mobile app for teachers and students.

9.3 Learning Outcomes and Career Relevance

The project exercises the main skills of the course: modelling data with forms and lookups, building reports and a dashboard, scripting with Deluge, configuring form workflows, setting up

e-mail notifications and a calendar workflow, configuring roles, permission profiles and data sharing, and deploying a version to Production. It also shows the importance of checking automation against real records, as the defects in Table 6.4 demonstrate. These skills are relevant to roles in business-application development, ERP and workflow automation, and low-code or citizen-development support.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

The checklist reflects the supplied evidence. “Partial” marks features that are only partly evidenced. Examiners should verify the entries against the live application.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created and sections / menu organised	Y	Menu: Student Details, Assignment Details, Submission Details, dashboard	5.1	Figures 4.6 and 5.8
II	Forms with ≥ 8 distinct field types	Y	Single Line, Drop Down, Email, Phone, Date, Number, Multi Line, Lookup (types read from the forms)	5.2	Figures 5.2 to 5.5
II	Lookup field (relationship between forms)	Y	Assignment → Subject; Submission → Student and Assignment	4.2, 5.2	Figures 4.2 and 5.4
II	Subform (parent–child data)	N	Not used	–	–
II	Field validations / layout rules	Y	Mandatory Department; Validate Submission Details workflow; layout rules not evidenced	5.2, 5.5	Tables 4.2 to 4.5; Code 5.1
II	List report with filter, grouping, custom action	Partial	Student, subject, assignment and submission list reports; grouping and custom actions not evidenced	5.3	Figures 5.6 and 5.7
II	Kanban / Calendar / Timeline / Map report	N	Not used	5.3	–
II	Pivot table / chart report	N	Not used	5.3	–
II	Page / dashboard built with UI Builder	Y	Dashboard with count tiles, pie chart and buttons	5.4	Figure 5.8
II	Deluge – form events (validate / success / field actions)	Y	Validations on form submission; Successful form submission	5.5, 5.6	Figures 5.9 and 5.10; Appendix C
II	Deluge – functions, fetch, loops, collections	Partial	Fetch with criteria in Check Late Submission; no loops or functions	5.5	Code 5.2
II	E-mail / in-app notifications	Y	Administrator e-mail and student e-mail	5.5, 5.7	Figures 5.12 and 5.13
II	Form workflow	Y	Four on Submission Details and one on Assignment Details	5.6	Figures 5.9 and 5.10
II	Scheduled workflow	N	Not used	5.6	–
II	Approval process	N	Not used	5.6	–
II	Blueprint	N	Not used	5.6	–
II	Batch / report workflow	N	Not used	5.6	–
II	Testing and debugging	Y	Test evidence table and observations	6	Tables 6.1 and 6.3; Figures 6.1 and 6.2
III	Integration with a Zoho service	Partial	Zoho Calendar workflow configured; event not verified	5.7	Figures 5.10 and 5.11
III	Connection / REST API call (invokeurl)	N	No invokeurl call evidenced	5.7	–
III	Webhook or Creator REST API	N	Not used	5.7	–

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Users, roles and permission profiles	Y	4 roles, 3 profiles, 2 data sharing rules, 1 user	5.8, 7.3	Figures 5.14 to 5.17
III	Customer portal	Partial	Portal address and sign-in page designed; no portal user tested	5.11, 7.1	Figure 5.18
III	Record-level access criteria	Partial	Described in the profile descriptions and sharing rules	5.8, 7.3	Figures 5.15 and 5.16
III	Zia / AI feature	N	Not used	5.9	–
III	Analytics (charts / Zoho Analytics)	Partial	Native dashboard pie chart; Zoho Analytics not used	5.9	Figure 5.8
IV	Performance optimisation applied and measured	N	Proposals only	7.2	Table 7.2
IV	Deployment across environments	Y	Published to Production (version 2.0)	7.1	Figure 5.1
IV	Mobile app / mobile layout	N	Preview icons only	5.11	–
IV	Pro-code extension	N	Not used	5.10	–

APPENDIX B – COURSE OUTCOME MAPPING

The self-assessed and examiner levels are left blank for completion at the time of evaluation.

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	[ ]	[ ]
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	[ ]	[ ]
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9, 7.3	[ ]	[ ]
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	[ ]	[ ]
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	[ ]	[ ]

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Validate Submission Details (Submission Details → Created or Edited → Validations on form submission)

```
// Script ID : DS-01
// Location  : Submission Details form -> Created or Edited
//             -> Validations on form submission -> Validate Submission Details
// Purpose   : Cancel the submission if the assignment, student or status is empty
if(input.Assignment.isNull())
{
    alert "Please select an assignment.";
    cancel submit;
}
if(input.Student.isNull())
{
    alert "Please select a student.";
    cancel submit;
}
if(input.Submission_Status.isNull())
{
    alert "Please select a submission status.";
    cancel submit;
}
```

C.2 Check Late Submission (Submission Details → Created or Edited → Validations on form submission)

```
// Script ID : DS-02
// Location  : Submission Details form -> Created or Edited
//             -> Validations on form submission -> Check Late Submission
// Purpose   : Set the status to Late or Submitted by comparing the submission
//             date with the due date of the selected assignment
if(!input.Assignment.isNull() && !input.Submission_Date.isNull())
{
    assignmentRecord = Assignment_Details[ID == input.Assignment];
    if(assignmentRecord.count() > 0)
    {
        if(input.Submission_Date > assignmentRecord.Due_Date)
        {
            submission_status = "Late";
        }
        else
        {
            submission_status = "Submitted";
        }
    }
}
```

C.3 Scripts Whose Source Was Not Captured

The bodies of three scripts were not visible in the supplied screenshots and are therefore not reproduced: DS-03 Student Status Update Notification, DS-04 Admin Submission Notification and DS-05 Calendar (Assignment Details → Successful form submission). Their locations and purposes are given in Table 5.5. Their source should be copied from the Deluge editor into this appendix before the report is submitted.

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the application portal at https://pavitham2024csbsrajal141.zohocreatorportal.in. Users sign in with the e-mail address of their account and choose Next, or choose Sign Up to create an account. The menu on the left shows Student Details, Assignment Details, Submission Details and the dashboard. The application can be opened in a desktop, tablet or phone browser; the Zoho Creator mobile app has not been configured.

D.2 Instructions for the Student

1. Sign in to the portal.
2. Open Assignment Details → Assignment Details Report to see the available assignments, their subjects and due dates.
3. Open Submission Details → Submission Details.
4. Select your name in the Student list and the assignment in the Assignment list.
5. Enter the Submission Date and select the Submission Status, then choose Submit.
6. Watch for the e-mail "Assignment Submission Status Updated" that tells you the status of your submission, and open the Submission Details report to read your marks and feedback.

D.3 Instructions for the Teacher

1. Sign in and open Assignment Details → Assignment Details to create an assignment: enter the title, choose the subject, and enter the description, assigned date, due date and maximum marks.
2. Open Submission Details → All Submission Details to see the submissions.
3. Open a submission, change the Submission Status (for example to Evaluated), enter the Marks Obtained and Feedback, and save. The student is e-mailed.
4. Open the dashboard to see the counts of students and subjects and use the buttons to submit an assignment or update a submission status.

D.4 Instructions for the Administrator

1. Open Student Details → Student Details and enter the Student ID, name, e-mail address, department (mandatory) and phone number, then submit.
2. Open the second Student Details section → Subject Details and enter the Subject ID, name, faculty name and semester, then submit.
3. Open Student Details → student details report and the other reports to review the records.
4. Open Settings → User Permissions to review the roles, the permission profiles and the data sharing rules.

5. Open the console menu Users to add users and assign them to roles and profiles.
6. Open Deploy → Environments to publish a new version from Development to Production.